

Ext

8

Projectile Motion

In this chapter, we will learn about

- the motion of an object projected horizontally
- the motion of an object projected at an angle
- the trajectory of a projectile
- the energy change in projectile motion

Ext

8.1 Understanding projectile motion

Flipped classroom

Watch a video and answer the questions.

DSE goal

Describe the shape of the path taken by a projectile launched at an angle of projection.

Understand the independence of horizontal and vertical motion.

DSE exam

17(1A)Q9, 19(1A)Q13

1 Introduction to projectile motion

Consider a boy throwing a basketball (Fig 8.1a). Once the basketball leaves the boy's hands, it moves freely under gravity—it is subject only to gravitational force. In all the cases, the basketball performs **projectile motion** and it is called a **projectile**. In particular, the **trajectories** (path) ② and ③ of the basketball are in the shape of a **parabola**.



Fig 8.1a *A basketball performs projectile motion once it leaves the boy's hands.*

projectile motion 拋體運動　　projectile 拋體　　trajectory 軌跡　　parabola 拋物線

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In fact, there are many examples of projectile motion in daily life (Fig 8.1b).



Water performs projectile motion once it leaves the nozzle.



Sparks (burning pieces) ejected from fireworks perform projectile motion.

Fig 8.1b *Projectile motion in daily life.*

2 Explaining projectile motion with Newton's laws of motion

Unless otherwise specified, we shall neglect air resistance in the discussion in this chapter.

▶ The net force (the gravitational force, mg) acts vertically downwards on a projectile. By Newton's second law, a projectile moves with a constant acceleration g pointing vertically downwards (Fig 8.1c). Since the projectile does not experience any horizontal forces, by Newton's first law, its horizontal speed is constant.

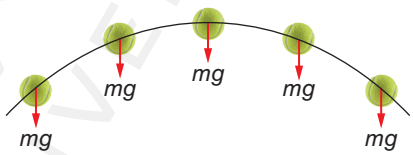


Fig 8.1c *A projectile is acted on only by the gravitational force mg.*

The above discussion is summarized in Table 8.1a. Note that the horizontal motion and vertical motion can be considered separately. They are independent of each other.

Although the basketball in Figure 8.1a on p.330 is acted on by the same net force in all cases, it moves along different trajectories. Can you explain why?

	Force on projectile	Motion of projectile
Vertical direction	mg (the net force)	moving with constant downward acceleration g
Horizontal direction	0	moving with constant speed

▶ ***Table 8.1a*** *Force and motion along vertical and horizontal directions in projectile motion.*

- **When air resistance is negligible, a projectile moves at a uniform speed horizontally and at a uniform acceleration vertically due to gravity.**
- **The horizontal motion and vertical motion are independent of each other.**

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3 Analysing horizontal and vertical motion of a projectile

We will further analyse the horizontal motion and vertical motion of a projectile in the following experiment.

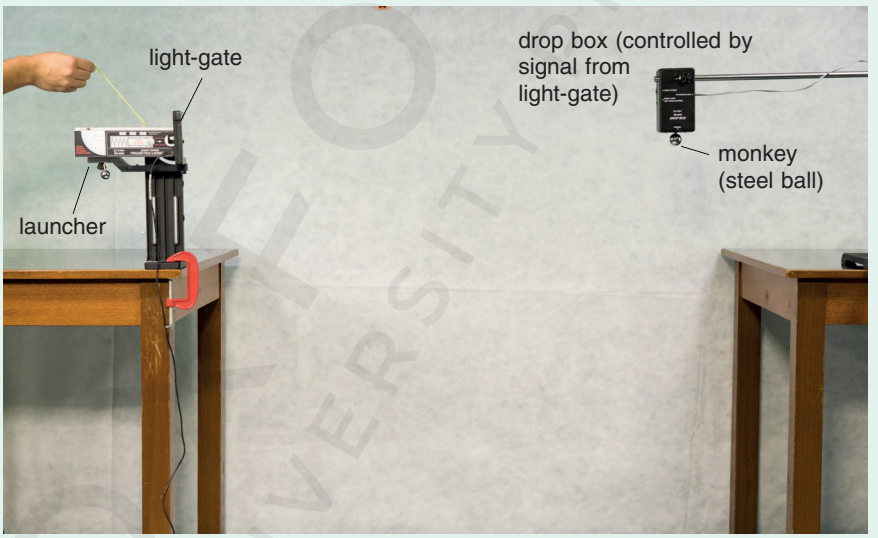
Video

→ This video demonstrates Expt 8a.

- Wear safety spectacles.
- Keep a safe distance from the apparatus.

Experiment 8a / Studying vertical motion of a projectile ('Monkey and hunter' experiment)

1 Set up the 'monkey and hunter' apparatus as shown. Ensure that the launcher is horizontal and that it is aimed at the monkey.



2 Launch a bullet. This switches off the magnetic attraction from the drop box. The monkey starts to fall at the instant the bullet is launched and passes the light-gate. Observe what happens.

3 Repeat step 2 by placing the monkey at other distances from the launcher.

Discussion

1 What kinds of motions do the bullet and the monkey perform?

2 Does the bullet always hit the monkey? What does this tell us about the vertical distances that they travel?

→ This is a Video Quiz about 'Monkey and hunter' experiment.

In Experiment 8a, the bullet hits the monkey. This means that the bullet travels the same vertical distance as the monkey does. Their motion can be further analysed in the *stroboscopic photo* (Fig 8.1d on p.333). It can show the positions of a moving object at **constant time interval**.

stroboscopic photo 頻閃照片

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A stroboscopic photo is taken in one single long exposure with light flashing at constant time intervals. It is similar to a multiple exposure photo that is obtained by taking a number of photos and combining them into one.

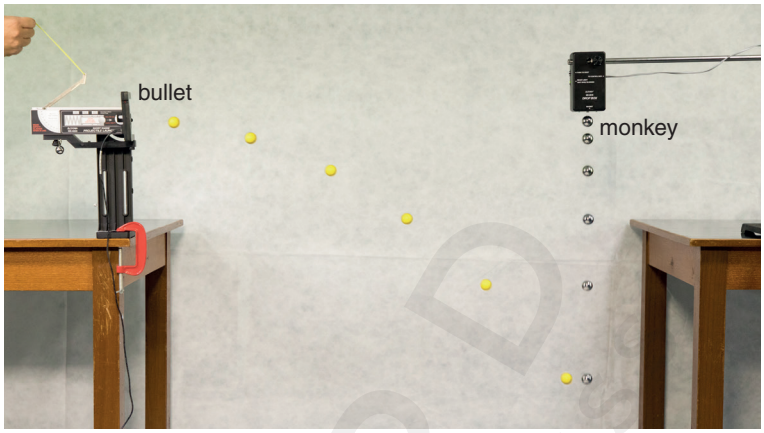


Fig 8.1d *A stroboscopic photo showing the positions of bullet and monkey at constant time interval.*

From the photo, we can find that, within any certain time interval, the vertical distance travelled by the bullet is the same as that travelled by the monkey. The result is independent of the initial distance between the launcher and the monkey (Fig 8.1e). This shows that the **vertical motion** of a projectile is a **uniformly accelerated motion under gravity**.

The monkey performs uniformly accelerated motion under gravity.

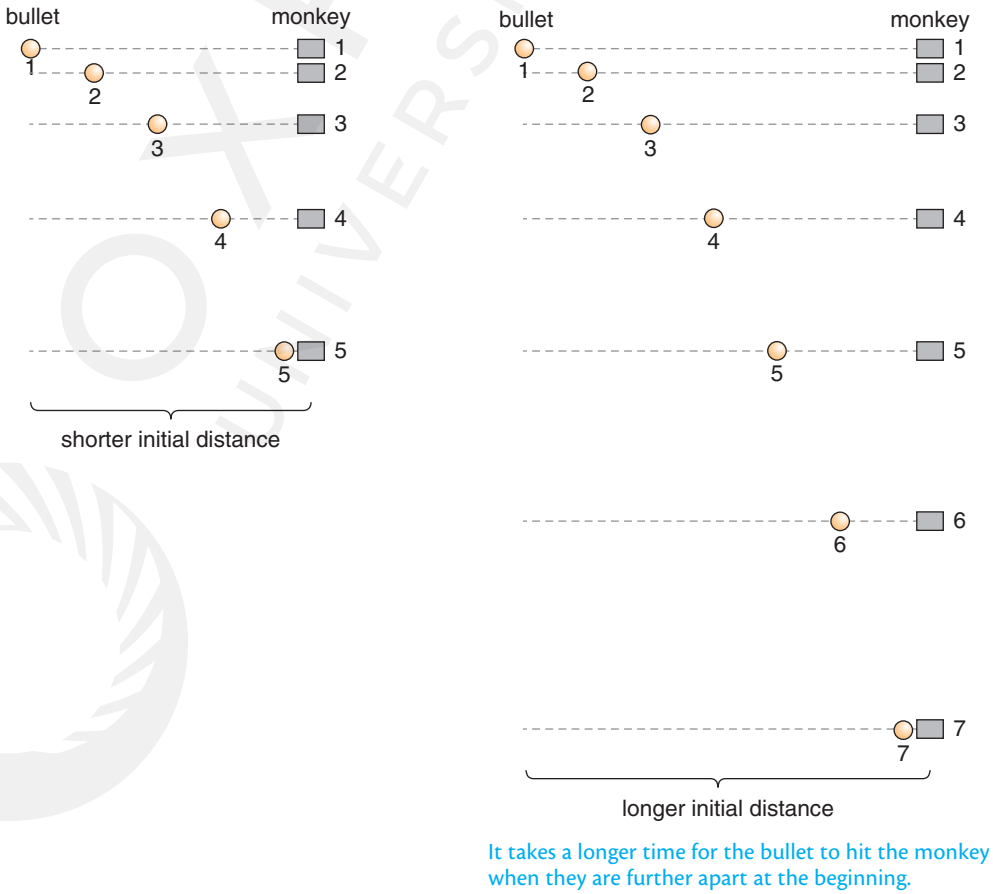


Fig 8.1e *The bullet and the monkey travel the same vertical distance within the same time interval (the numbers help you match the images of the bullet and the monkey taken at the same instant).*

Ext

Simulation

→ This simulation allows Ss to project two balls horizontally from different vertical positions at the same time and observe their horizontal motions. It is available on OUP web in PC version (.exe) only.

→ This is a Video Quiz about the experiment 'A ball projected horizontally'.

Moreover, from the photo, we can see that the bullet travels the same horizontal distance within every time interval (Fig 8.1f). This means the horizontal speed of the bullet remains the same throughout the projectile motion. In other words, the **horizontal motion** of a projectile is a **uniform motion**.

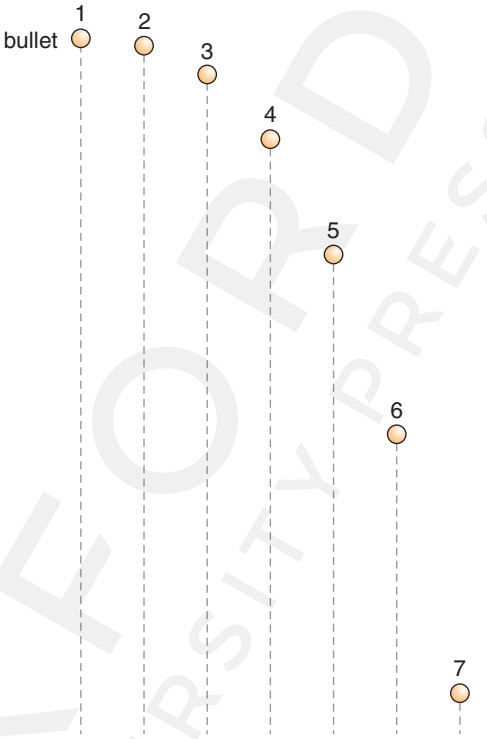


Fig 8.1f *The bullet travels the same horizontal distance within every time interval.*

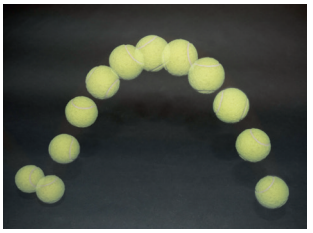


Fig 8.1g *A stroboscopic photo showing the projectile motion of a ball.*

In fact, the projectile motion (Fig 8.1g) can be regarded as a combination of uniformly accelerated vertical motion and uniform horizontal motion (Fig 8.1h).

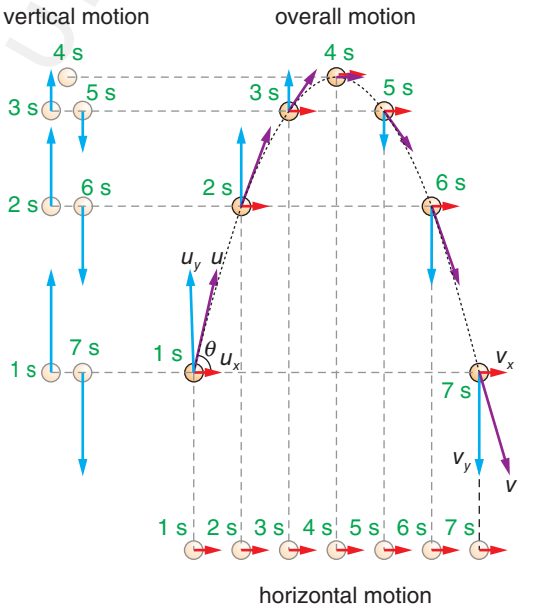


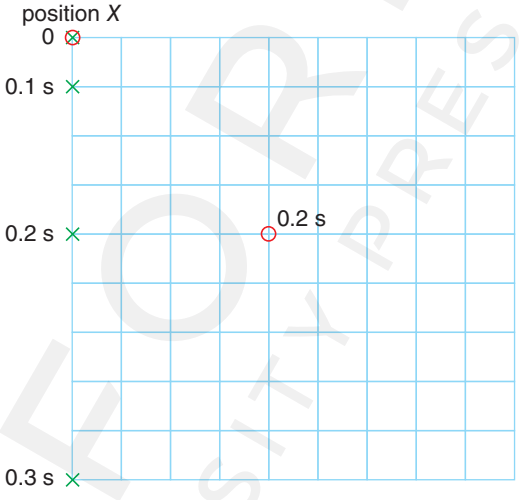
Fig 8.1h *Regard projectile motion as a combination of the vertical and horizontal motion.*

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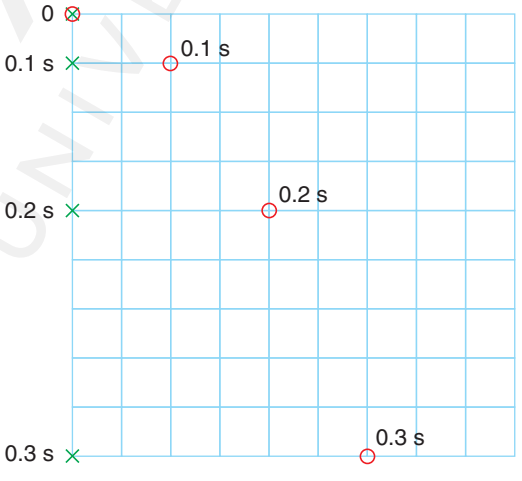
Example 1 Tracing the position of a projectile

Tennis ball P falls freely from rest at X at time $t = 0$. The crosses (×) in the figure show the positions of P at $t = 0$, 0.1 s, 0.2 s and 0.3 s.

Another tennis ball Q is projected horizontally from X at $t = 0$. The circle (○) in the figure shows the positions of Q at $t = 0$ and 0.2 s. In the figure, mark the positions of Q at $t = 0.1$ s and 0.3 s. Neglect air resistance.



Solution



Tip

Q moves uniformly in the horizontal direction. As it moves 4 grids in the horizontal direction in 0.2 s, it moves 2 grids in every 0.1 s.

On the other hand, the vertical motion of Q is the same as P. Starting at the same height, the height of Q at any instant is always the same as P.

Think about...

Do balls P and Q have the same speed at $t = 0.2$ s?

Now, let us try the activity on p.336. See if you can explain the results by applying the knowledge that you acquire in this section.

Ext

Projectile motion

Phone experiment

Part A: Comparing vertical motion Video → This video demonstrates the Phone experiment (Part A).

What to do

1 Place coin *X* and a ruler at the edge of a table. Place coin *Y* on the ruler. You will let the coins start to fall at the same time.

2 Switch on the camera app of your smartphone. Fix the phone and make sure its view-of-field covers the ground below them. Start taking a slow motion video (Fig a).

3 Flick the end of the ruler (Fig b). *X* is projected horizontally and *Y* falls vertically. Observe their motion from the video taken.

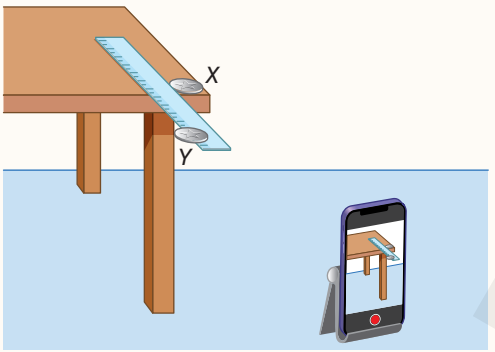


Fig a

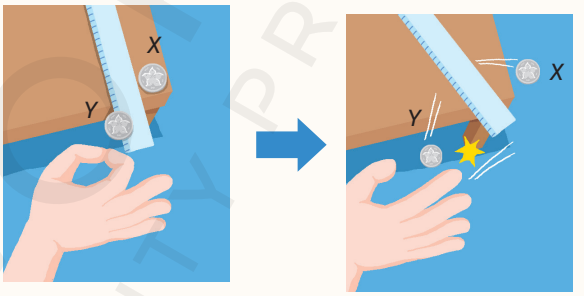


Fig b

What you find

Is there any difference between the vertical motion of the two coins?

Part B: Comparing horizontal motion Video → This video demonstrates the Phone experiment (Part B).

What to do

1 Cut an empty kitchen paper roll in half lengthways. Using two drink cartons as stands, raise one end of each half-roll to make two identical inclined tracks *P* and *Q*. Place *P* on a table and place *Q* on the ground vertically below *P*.

2 Repeat step 2 in part A (Fig c).

3 Release two stationary ping-pong balls from the upper ends of *P* and *Q* at the same time (Fig d). Observe the balls' motion from the video taken. Change the vertical distance between the tracks and repeat the above steps.

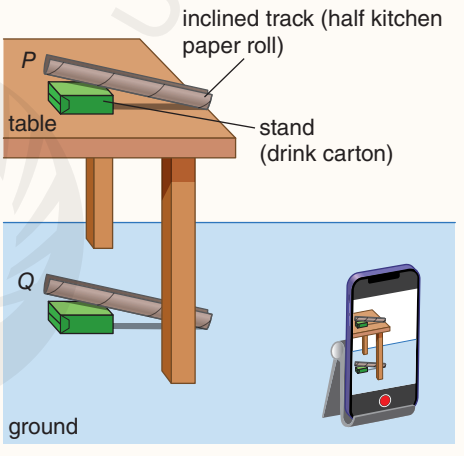


Fig c

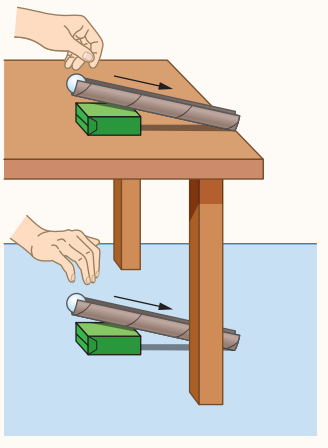


Fig d

What you find

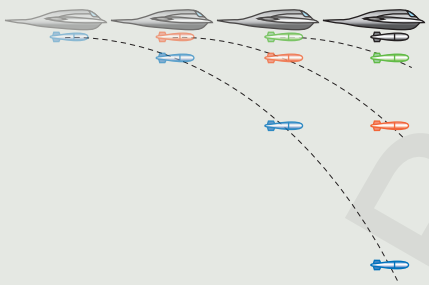
Is there any difference between the horizontal motion of the two balls?

Ext

Know more

Alignment of bombs dropped by a bomber

When a bomber flies horizontally at a constant velocity and drops bombs one by one, the bombs undergo projectile motion with the same horizontal speed as the bomber. As a result, they always stay underneath the bomber and align vertically.



Each bomb moves in a projectile motion with the same horizontal speed as the bomber.



Alignment of bombs dropped by a bomber at different instants.

Google Forms Microsoft Forms

Quick check 1

→ E-version in OUP Exercise Platform is also available.

1 Stone *A* is thrown horizontally at speed *v* from a cliff and stone *B* is dropped from rest from the cliff at the same moment. Assume the air resistance is negligible. Which of the following statements is/are correct? Put a tick in the corresponding box(es).

- ☐ *B* reaches the surface of the water first.
- ☐ *A* and *B* have the same acceleration during their flights.
- ☐ *A* and *B* have the same vertical speed when they reach the surface of the water.
- ☐ The horizontal speed of *A* keeps decreasing during its flight.

2 A bird flies horizontally at a constant velocity (Fig a). If it drops a seed when it is vertically above *B*, who will most probably be hit?

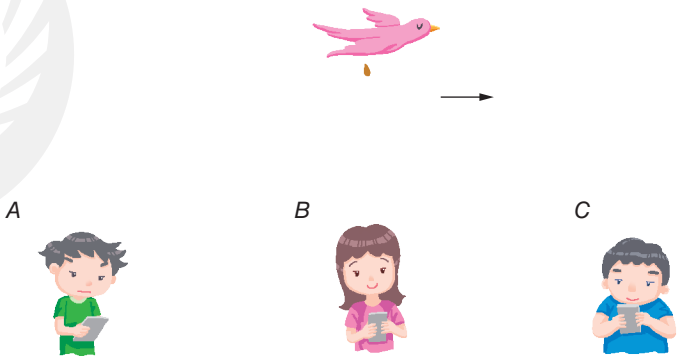


Fig a

Ext

Exercise 8.1

Neglect air resistance.

1 Which of the following statements about a projectile is **incorrect**?

A Its weight is the only force that acts on it.

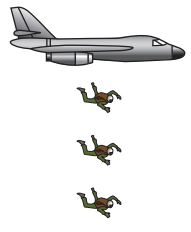
B It moves at a constant horizontal speed.

C It moves at a constant vertical speed.

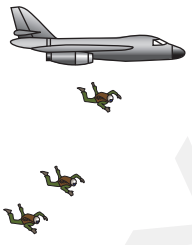
D Its vertical and horizontal motions are independent of each other.

***2** Three skydivers leave with zero vertical velocity at one-second intervals from a plane flying horizontally at a constant velocity. Which of the following diagrams best shows the positions of the plane and the skydivers at a certain instant?

A



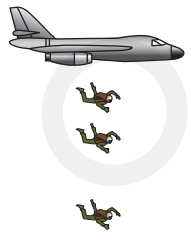
B



C



D



***3** P and Q are two identical coins. At the instant shown (Fig a), P is released to fall vertically and Q is projected horizontally from the same height.

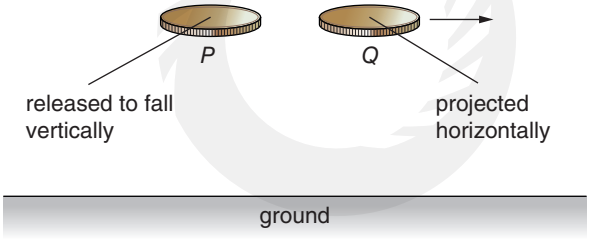


Fig a

Which of the following descriptions about P and Q is/are correct?

(1) They reach the ground at the same time.

(2) They have the same vertical velocity when they reach the ground.

(3) They have the same horizontal velocity when they reach the ground.

A (1) only

B (1) and (2) only

C (2) and (3) only

D (1), (2) and (3)

***4** An object A is dropped vertically from rest at time $t = 0$. Its positions at every second are shown in orange in Figure b. Another object B is projected horizontally with a certain speed u. X and Y are its positions at $t = 0$ and 1 s respectively.

(a) Draw the positions of B at $t = 2$ s and 3 s in Figure b.

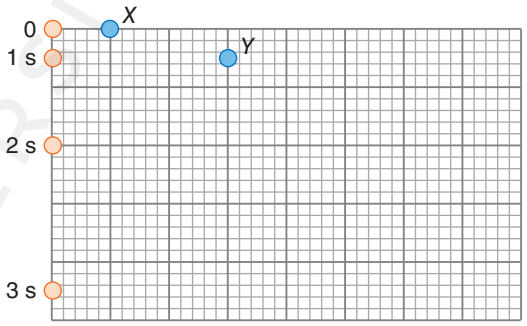


Fig b

(b) If B is projected from X horizontally at speed $\frac{u}{2}$, what will its position be at $t = 1$ s, 2 s and 3 s? Draw the positions in Figure c.

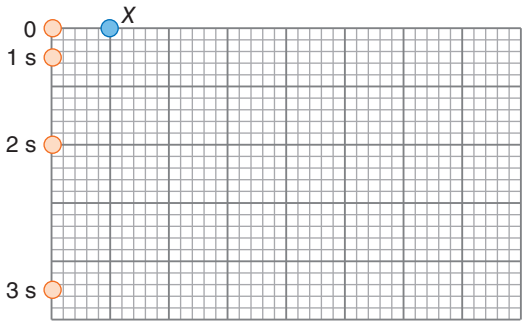


Fig c

Ext

8.2 Analysing projectile motion

DSE goal

Solve problems involving projectile motion.

DSE exam

12(1A)Q12, 14(1A)Q10, 16(1A)Q10, 17(1B)Q4(a), 22(1A)Q13

1 Projectile launched horizontally

To solve problems involving projectile motion, we may consider the horizontal and vertical motion separately. First, let us recall case ② on p.330, in which a basketball is projected horizontally to the right with speed u at time $t = 0$ (Fig 8.2a).

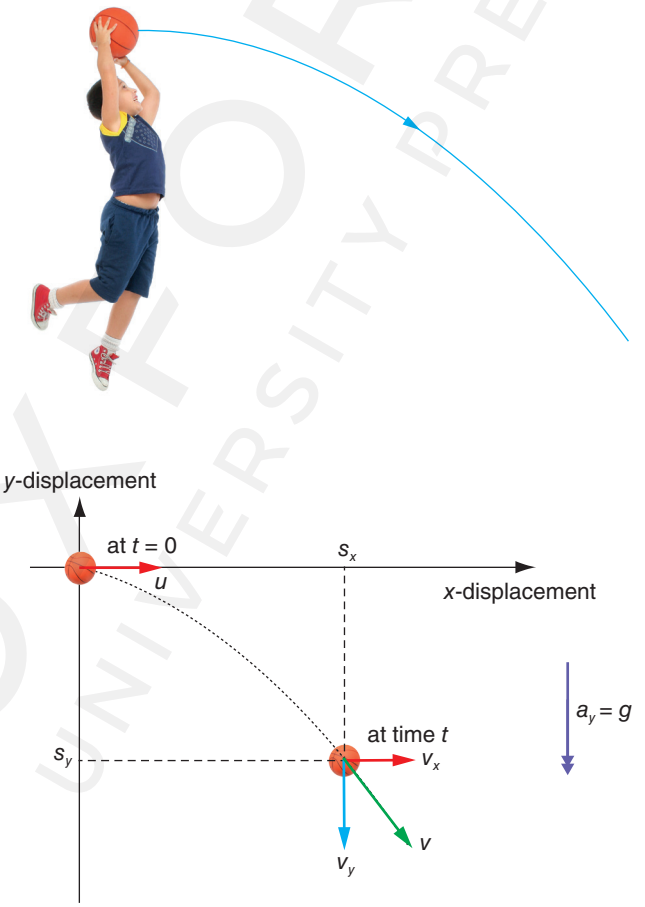


Fig 8.2a *The displacement and velocity of a basketball projected horizontally.*

We know that:

- along the horizontal direction x (right as positive),

  initial velocity $u_x = u$

  acceleration $a_x = 0$

- along the vertical direction y (downwards as positive),

  initial velocity $u_y = 0$

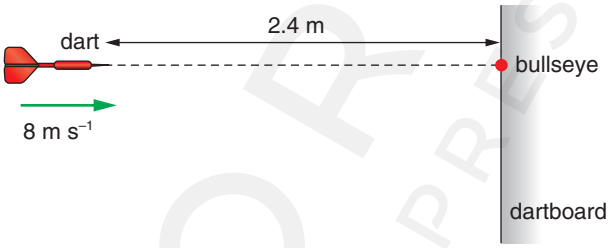
  acceleration $a_y = g$

Ext

The following example shows how we can solve problems by applying Newton's laws to the horizontal motion and vertical motion.

Example 2 Horizontal and vertical distances travelled

Gareth is 2.4 m away from a dartboard. He throws a dart horizontally with a speed of 8 m s^{-1}. Neglect air resistance.



(a) Find the time needed for the dart to hit the dartboard.

(b) The dart is launched when it is at the same level as the bullseye. Determine where on the dartboard the dart would hit.

(c) 'If Gareth throws the dart horizontally, he will have to throw it at a higher level in order to hit the bullseye.' Explain the statement.

Solution

(a) Consider the horizontal direction.

Tip
Apply $s_x = u_x t$.

$$\text{Time needed} = \frac{s_x}{u_x}$$
$$= \frac{2.4}{8}$$
$$= 0.3 \text{ s}$$

(b) Take the downward direction as positive.
Consider the vertical direction.

Tip
From (a), the dart takes 0.3 s from launching to hitting the dartboard. In this journey, $u_y = 0$ and $a_y = 9.81$ m s^{-2}.

$$s_y = u_y t + \frac{1}{2} a_y t^2$$
$$= 0 + \frac{1}{2} \times 9.81 \times 0.3^2$$
$$= 0.441 \text{ m}$$

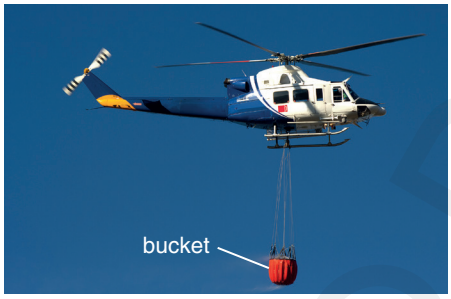
$\therefore$ The dart hits a point 0.441 m below the bullseye.

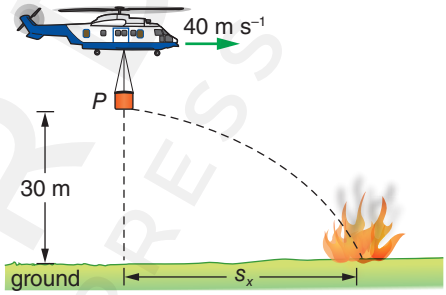
(c) The dart falls under gravity, so it always reaches the dartboard at a position lower than its initial position. Therefore, the dart should be thrown at a position higher than the bullseye.

Ext

Example 3 Final speed of a projectile

A fire breaks out in a remote area. A helicopter carrying a bucket of water flies horizontally at a constant velocity of 40 m s^{-1} towards the right. The bucket is 30 m above the ground. Water is released when the bucket reaches position P. The water then falls onto the fire. Neglect air resistance.





(a) Where is the helicopter when the water reaches the fire?

(b) Find the horizontal distance s_x between P and the fire.

(c) Find the speed v of the water when it reaches the fire

Solution

Tip
The helicopter and the water have the same horizontal speed.

(a) Directly above the fire

(b) Take the downward direction as positive.

Consider the vertical direction.

$$s_y = u_y t + \frac{1}{2} a_y t^2$$

$$30 = 0 + \frac{1}{2} \times 9.81 \times t^2$$

$$t = 2.473 \text{ s}$$

Tip
It takes time t for the water to travel a vertical distance of 30 m; it also takes time t for the water to travel a horizontal distance s_x.

Consider the horizontal direction.

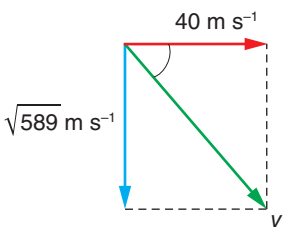
$$s_x = u_x t = 40 \times 2.473 = 98.9 \text{ m}$$

(c) Consider the vertical direction.

$$v_y^2 = u_y^2 + 2a_y s_y = 0 + 2 \times 9.81 \times 30 = 589$$

$$v_y = \sqrt{589} \text{ m s}^{-1}$$

$$v = \sqrt{v_x^2 + v_y^2} = \sqrt{40^2 + 589} = 46.8 \text{ m s}^{-1}$$



Think about...

Should the water be released earlier or later if

(a) the helicopter flies at a smaller horizontal velocity;

(b) the bucket is hung at a higher position?

Ext

Google Forms Microsoft Forms

Quick check 2

→ E-version in OUP Exercise Platform is also available.

Neglect air resistance.

1 A ball is thrown horizontally from position X at speed v and hits the ground after time t. X is at a height h above the ground. Does t depend on v and h?

2 A ball is thrown horizontally from position Y at speed v and hits a wall in front of it after time t. Y is at a height h above the ground and at a horizontal distance d from the wall. Does t depend on v, h and d?

3 Balls X and Y are projected horizontally at speeds u_X and u_Y respectively. Both of them hit point R (Fig a).

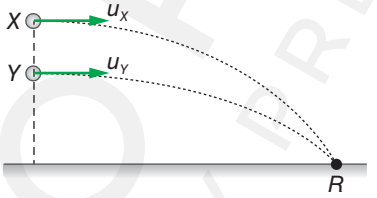


Fig a

Which of the following comparisons is/are correct? Put a tick in the corresponding box(es).

- ☐ X and Y spend the same time for their flights.
- ☐ X and Y have the same acceleration.
- ☐ X has a higher vertical speed when hitting the ground.
- ☐ Y has a higher horizontal speed when hitting the ground.
- ☐ u_X equals u_Y.

Learn by practice 1

1 A ball is thrown horizontally as shown (Fig a).

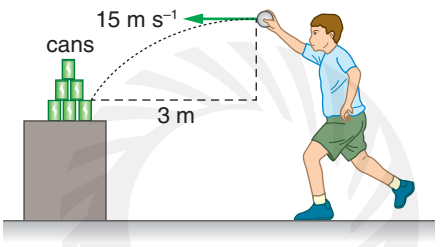


Fig a

(a) Find the time it takes to hit the cans.

(b) Find the vertical distance travelled by the ball in the period of time in (a).

2 A ball is thrown horizontally with a speed of 1.5 m s^{-1} at a certain height above the ground. If the ball hits the ground after travelling a horizontal distance of 1.4 m, what is its initial height from the ground?

3 A table tennis ball is hit and flies horizontally as shown (Fig b).

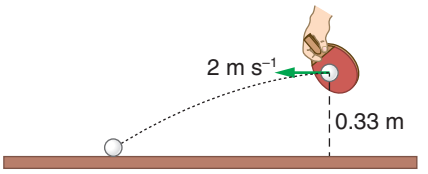


Fig b

(a) Find the time it takes to reach the table.

(b) Find the horizontal distance travelled by the ball in the period of time in (a).

4 A helicopter flies horizontally at 200 m above the surface of the water with a speed of 12 m s^{-1}. A package is dropped from the helicopter. Find the speed of the package when it reaches the surface of the water.

Ext

→ Start Oxford Physics XR 'Projectile motion' and scan this page to play.

iOS Android

2 Projectile launched at an angle

Pre-study **Resolving velocities**

Velocity is a vector. Like a force, velocity can also be resolved into two components perpendicular to each other.

Suppose a velocity vector $\vec{u}$ makes an angle θ with the horizontal (Fig a). $\vec{u}$ can be resolved into a horizontal component $\vec{u_x}$ and a vertical component $\vec{u_y}$ as follows (Fig b):

$u_x = u \cos\theta$ and $u_y = u \sin\theta$

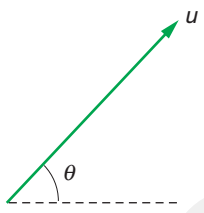


Fig b

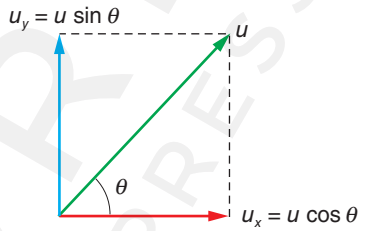


Fig b

DSE exam

12(1B)Q5, 13(1A)Q13, 15(1B)Q3, 18(1B)Q4, 19(1A)Q6, 20(1B)Q4, 21(1B)Q3, 23(1A)Q13

Consider an object projected with velocity u at an angle θ, i.e. the **angle of projection**, to the horizontal at time $t = 0$ (Fig 8.2b).

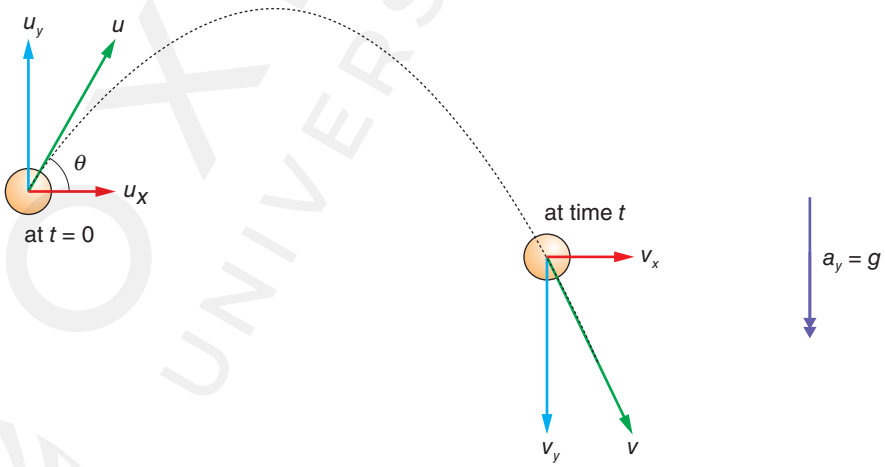


Fig 8.2b *The velocity of an object projected at an angle.*

We know that:

- along the horizontal direction x (right as positive),

  initial velocity $u_x = u \cos\theta$

  acceleration $a_x = 0$

- along the vertical direction y (upwards as positive),

  initial velocity $u_y = u \sin\theta$

  acceleration $a_y = -g$

angle of projection 投射角

Ext

Note that:

1 A projectile has the **same speed** (due to the conservation of energy) and makes the **same angle** with the horizontal at the **same height**, e.g. at the two points shown in Figure 8.2c.

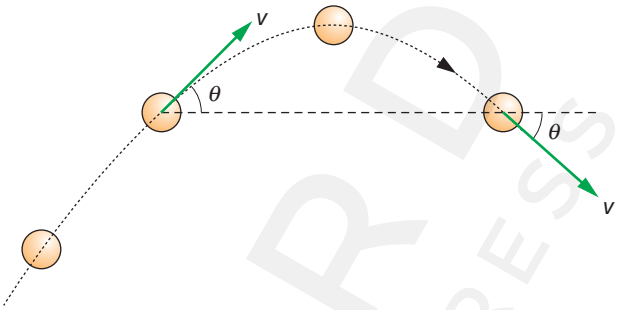


Fig 8.2c *Velocity v of a projectile at the same height.*

2 When a projectile is **at its maximum height** (Fig 8.2d), the vertical component of its velocity becomes zero (i.e. $v_y = 0$) but the horizontal component remains constant. Therefore, its velocity is not equal to zero (i.e. $v \neq 0$).

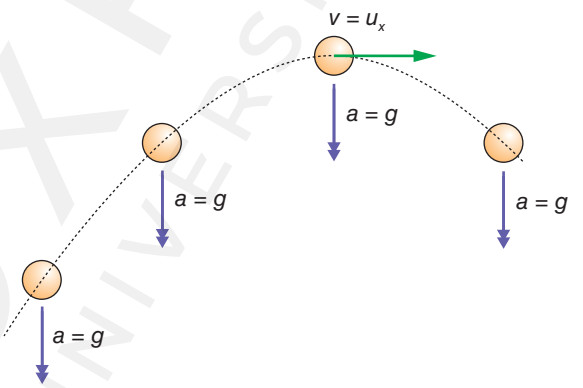


Fig 8.2d *At maximum height, v ≠ 0.*

Sometimes we may need to find the **time of flight** and the **range** of a projectile (Fig 8.2e). Their meanings are as follows:

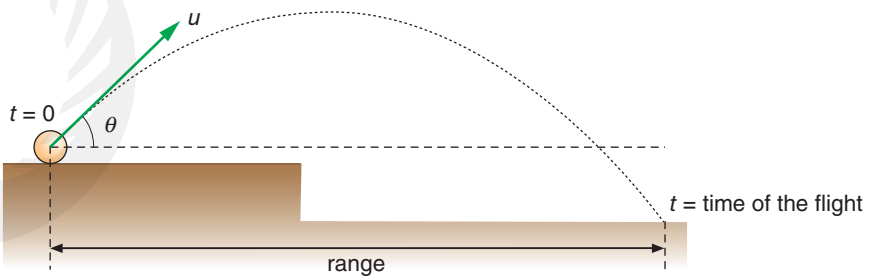


Fig 8.2e *Time of flight and range for projectile motion.*

1 Time of flight: the duration of time from projection to landing

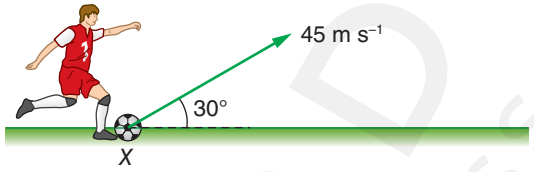
2 Range: the horizontal distance travelled from projection to landing

time of flight 飛行時間　　range 射程

Ext

Example 4 Time of flight and range

In a football game, a player kicks the football at X so that it flies off at an angle of projection of 30° with a speed of 45 m s^{-1}. Neglect air resistance.



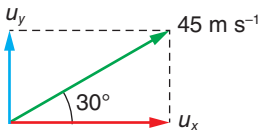
(a) Find the horizontal component u_x and vertical component u_y of the football's initial velocity.

(b) Find the maximum height that the football reaches.

(c) If the football lands at the same level as X, what is the time of flight and the range?

Solution

Take the upward direction as positive.



(a) $u_x = u \cos\theta = 45 \cos 30° = 39.0$ m s^{-1}

$u_y = u \sin\theta = 45 \sin 30° = 22.5$ m s^{-1}

(b) Consider the vertical direction.

$v_y^2 = u_y^2 + 2a_y s_y$

$0 = 22.5^2 + 2 \times (-9.81) \times s_y$

$s_y = 25.8$ m

The maximum height is 25.8 m.

Tip
$v_y = 0$ at maximum height.

(c) Consider the vertical direction.

$$s_y = u_y t + \frac{1}{2} a_y t^2$$

$$0 = 22.5t + \frac{1}{2} \times (-9.81) \times t^2$$

$t = 4.59$ s or 0 (rejected)

The time of flight is 4.59 s.

Consider the horizontal direction.

$s_x = u_x t = 39.0 \times 4.59 = 179$ m

The range is 179 m.

Think about...

If the football flies off with the same initial speed but a slightly larger angle of projection, how will the time of flight and the maximum height change?

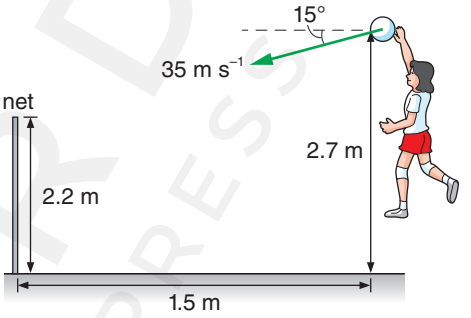
Tip
Should we consider the horizontal direction and/or the vertical direction when finding each of the two quantities?

Ext

Problems involving an angle of projection below the horizontal can be solved similarly. But we need to make sure the signs of the quantities are assigned correctly in applying the equations of motion.

Example 5 Angle of projection below the horizontal

Amy gives a smash in a volleyball match. She hits the ball at a position 2.7 m above the ground and 1.5 m away from the net. The ball leaves her hand at an angle of 15° below the horizontal with a speed of 35 m s^{-1}. If the net is 2.2 m high, will the ball cross the net? Neglect air resistance.

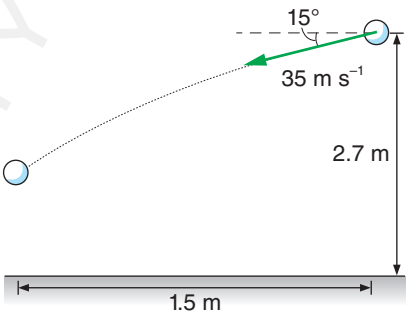


Solution

Take the downward direction as positive.

When the ball has travelled from Amy by a horizontal distance of 1.5 m,

$$t = \frac{s_x}{u_x} = \frac{1.5}{35 \cos 15°} = 0.0444 \text{ s}$$



Consider the vertical direction.

$$s_y = u_y t + \frac{1}{2} a_y t^2$$

$$= (35 \sin 15°) \times 0.0444 + \frac{1}{2} \times 9.81 \times 0.0444^2$$

$$= 0.412 \text{ m}$$

Height of the ball above the ground = 2.7 − 0.412 = 2.288 m > 2.2 m

∴ The ball will cross the net.

Quick check 3

→ E-version in OUP Exercise Platform is also available.

Google Forms Microsoft Forms

Neglect air resistance.

1 A ball is thrown at an angle above the horizontal from position X and hits a wall in front of it after time t (Fig a). The initial horizontal and vertical speeds of the ball is v_x and v_y respectively. Does t depend on h, d, v_x and v_y?

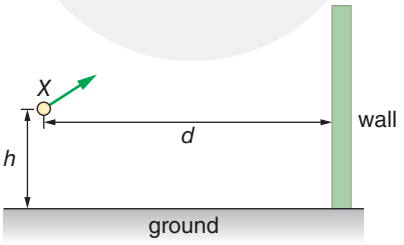


Fig a

2 A ball is thrown at an angle above the horizontal from position Y and hits the ground after time t. The initial horizontal and vertical speeds of the ball is v_x and v_y respectively. Does t depend on h, v_x and v_y?

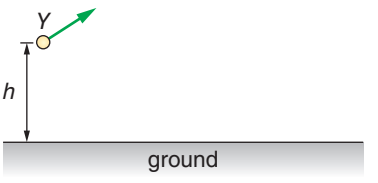


Fig b

Ext

Learn by practice 2

Neglect air resistance.

1 A ball is projected at an angle of 30° above the horizontal with a speed of 12 m s^{-1}. It hits the ground after some time. Find the time of flight and the range if the ball is projected from (Fig a)

(a) point X on the ground,

(b) point Y at a height of 2 m above the ground.

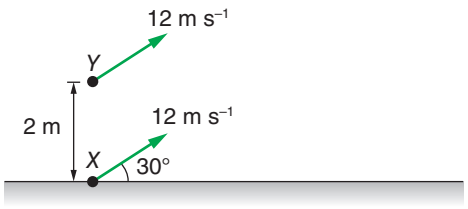


Fig a

2 A ball is projected at an angle of 35° above the horizontal with a speed of 10 m s^{-1} from a position 3 m above the ground. Find the maximum height above the ground that the ball reaches.

3 Peter throws a ball of mass 2 kg from the top of a cliff at a height of 50 m above the surface of the water. The ball is thrown with an initial speed of 5 m s^{-1} and at an angle of 15° below the horizontal.

(a) Find the time it takes for the ball to strike the surface of the water.

(b) Find the horizontal and vertical components of the ball's velocity when it strikes the surface of the water.

3 Projectile launched and landed at the same level

Simulation

→ This simulation allows Ss to freely choose the initial speed and the angle of projection of a cannonball. Ss can observe how the range is affected. It is available on OUP web in PC version (.exe) only.

In this part, we will examine a special case: a projectile that lands at the same level as its starting position (Fig 8.2f).

Its motion has the following properties.

1 The trajectory is symmetrical about the vertical line AB passing through the point of maximum height.

2 The time of upward flight is equal to the time of downward flight.

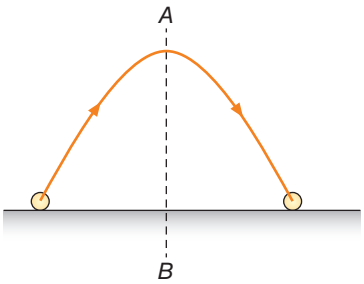


Fig 8.2f Projectile that launches and lands at the same level.

$u \uparrow$

$\Rightarrow u \sin\theta \uparrow$

$\Rightarrow$ maximum height $\uparrow$ and time of flight $T \uparrow$

$u \uparrow$ and $T \uparrow$

$\Rightarrow$ range $(= uT\cos\theta) \uparrow$

▶ **3** For a given angle of projection θ, the maximum height and the range increase as the launching speed u increases (Fig 8.2g).

→ This is a Video Quiz about the experiment 'Range and angle of projection'.

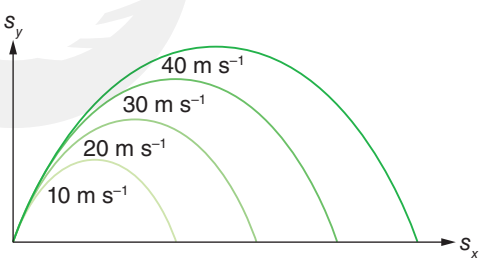


Fig 8.2g Trajectories with different u at the same θ.

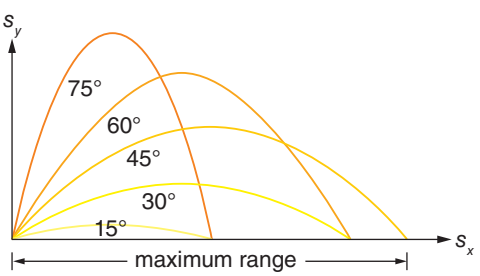


Fig 8.2h Trajectories with different θ at the same u.

Ext

$\theta \uparrow$
$\Rightarrow u \sin\theta$
$\Rightarrow$ maximum height $\uparrow$

Remind Ss that if the projectile does not land at the same height that it is launched from, the range may not be the maximum when $\theta = 45°$.

If $\theta_2 = 90° - \theta_1$, the projectile has the same range for θ_1 and θ_2.

▶ **4** For a given launching speed u, the maximum height increases as the angle of projection θ increases. The range also depends on θ but in a way as shown in Figure 8.2h on p.347.

- The range increases with θ from 0° to 45° and decreases with θ from 45° to 90°.
- When $\theta = 45°$, the range is maximum.
- ▶ There are two possible values of θ for a projectile to have the same range, except at 45°.

Know more

Projectile with air resistance

How will the projectile and its trajectory be affected if there is air resistance?

▶ p.366

Proofs for some properties in projectile motion

Properties in projectile motion may be proved mathematically. See the proofs on the pages indicated below.

▶ p.366–367

Google Forms Microsoft Forms

Quick check 4

→ E-version in OUP Exercise Platform is also available.

1 For a projectile that lands at the same level as the starting position, its range is the greatest when the angle of projection is _____°.

2 For a projectile that lands at the same level as the starting position, the maximum height that it reaches is the greatest when the angle of projection is _____°.

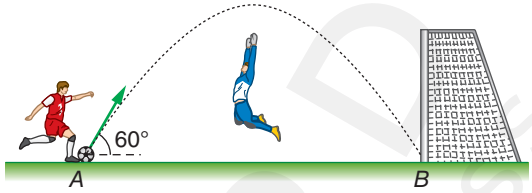
3 For a projectile that lands at the same level as the starting position, which of the following statements about its upward flight and its downward flight is/are correct? Put a tick in the corresponding box(es).

- ☐ The distance travelled during the upward flight is equal to the distance travelled during the downward flight.
- ☐ The time of the upward flight is equal to the time of the downward flight.
- ☐ The projectile moves at the same constant vertical speed during the upward flight and the downward flight.
- ☐ The projectile moves at the same constant horizontal speed during the upward flight and the downward flight.

Ext

Example 6 Properties of symmetrical trajectory

Teddy lobs a ball over the goalkeeper. The ball's time of flight from A to B is 1.4 s and its angle of projection is 60°. Neglect air resistance.



(a) Find the initial speed u of the ball.

(b) The goalkeeper is at a position directly below the maximum point reached by the ball. Find the horizontal distance between the goalkeeper and A.

(c) State the velocity of the ball when it reaches the ground.

(d) If Teddy lobs the ball at an angle of projection of 30°, the ball will also head towards B. State one reason for Teddy not to choose this angle.

Solution

Take the upward direction as positive.

(a) Consider the vertical motion of the ball from A to B.

$$s_y = u_y t + \frac{1}{2} a_y t^2$$

$$0 = u_y \times 1.4 + \frac{1}{2} \times (-9.81) \times 1.4^2$$

$$u_y = 6.867 \text{ m s}^{-1}$$

By $u_y = u \sin \theta$,

$$u = \frac{u_y}{\sin \theta} = \frac{6.867}{\sin 60°} = 7.93 \text{ m s}^{-1}$$

(b) Time needed to reach the maximum height $= \frac{1.4}{2} = 0.7$ s

Consider the horizontal motion of the ball from A to the maximum height.

$s_x = u_x t = u \cos \theta \times t = 7.93 \cos 60° \times 0.7 = 2.78$ m

The horizontal distance between the goalkeeper and A is 2.78 m.

(c) 7.93 m s^{-1} (towards right, 60° below horizontal)

(d) If the ball flies at an angle of projection of 30°, it will reach a smaller maximum height. It will be easier for the goalkeeper to catch the ball.

Ext

4 Energy conversion in projectile motion

The total mechanical energy of a projectile is constant during its flight if air resistance is negligible. Figure 8.2i shows the energy conversion in a projectile with launching speed u and maximum height H during its flight. The energy–time graph in Figure 8.2j also describes the change in different types of energy in the projectile.

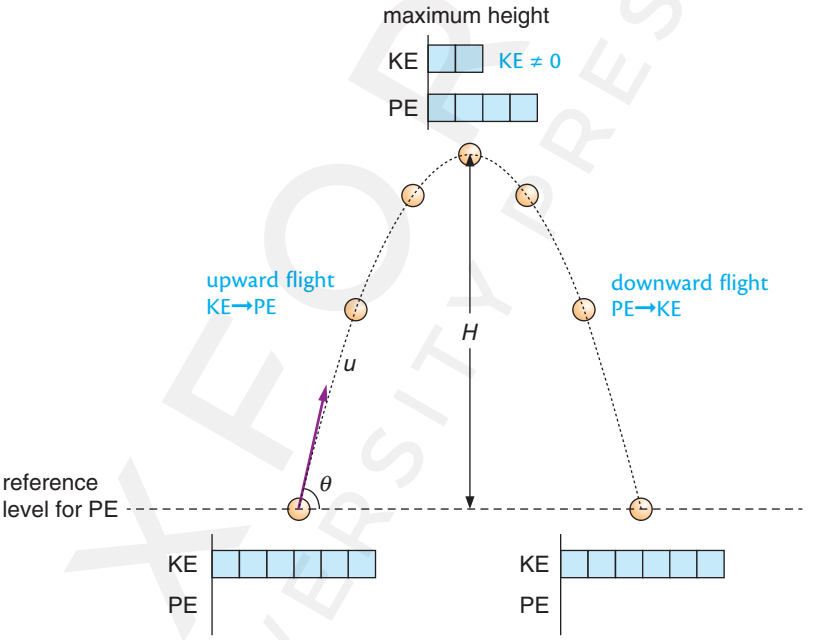


Fig 8.2i *Energy conversion of a projectile, taking the PE at the launching level as zero.*

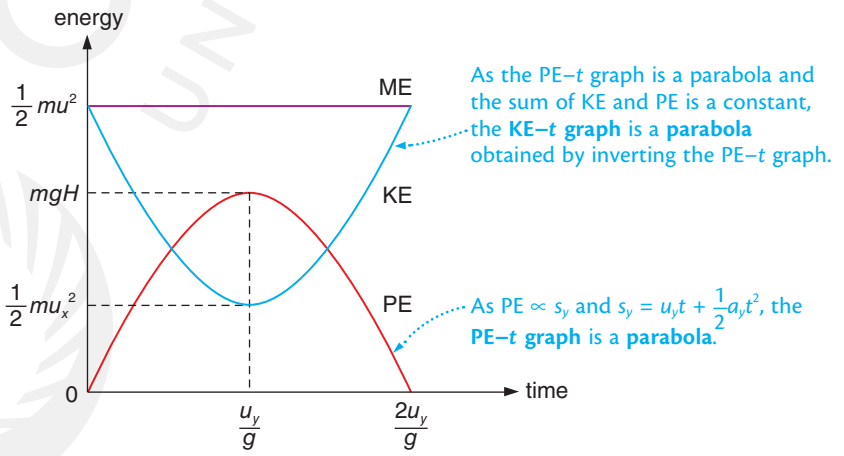


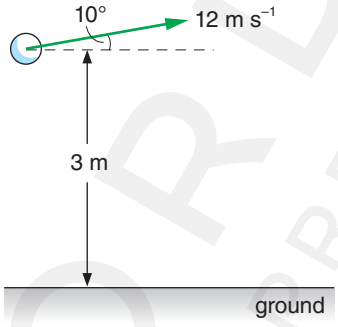
Fig 8.2j *Energy change of a projectile.*

Some problems can be solved by considering the energy conversion during the flight. When the energy approach is used, we do not need to consider the horizontal motion and vertical motion separately, as shown in Example 7.

Ext

Example 7 Analysing projectile motion using law of conservation of energy

An object is at a height 3 m above the ground. It is projected at 12 m s^{-1} at an angle of 10° above the horizontal and hits the ground. Neglect air resistance.



(a) Find the maximum height above the ground that the object reaches.

(b) Find the speed v of the object when it hits the ground.

Solution

Take the PE at the ground as zero.

(a) Consider the motion from the starting position to the maximum height.

$$KE_1 + PE_1 = KE_2 + PE_2$$

$$\frac{1}{2}mv_1^2 + mgh_1 = \frac{1}{2}mv_2^2 + mgh_2$$

$$\frac{1}{2} \times 12^2 + 9.81 \times 3 = \frac{1}{2} \times (12 \cos 10°)^2 + 9.81 \times h_2$$

$$h_2 = 3.22 \text{ m}$$

The maximum height above the ground is 3.22 m.

Tip
At maximum height, $v = u_x = u \cos \theta$

(b) Consider the motion from the starting position to the ground.

$$KE_1 + PE_1 = KE_2 + PE_2$$

$$\frac{1}{2}mv_1^2 + mgh_1 = \frac{1}{2}mv_2^2 + 0$$

$$\frac{1}{2} \times 12^2 + 9.81 \times 3 = \frac{1}{2} \times v^2$$

$$v = 14.2 \text{ m s}^{-1}$$

Note that θ does not appear in this equation. This means that the final speed depends on the initial speed and the height, but not the angle of projection.

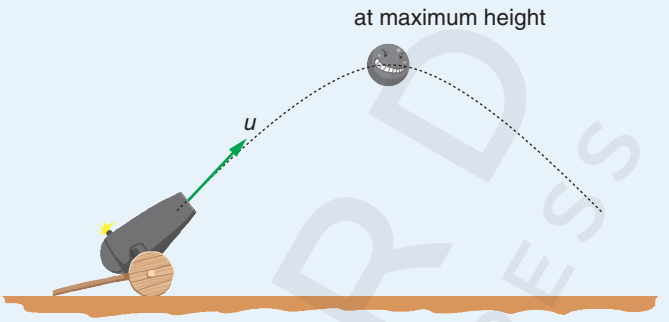
Think about...

Solve the above question using Newton's laws of motion. Can you obtain the same results?

Ext

Right and wrong

An object is projected upwards with speed u at an angle.



When the object reaches its maximum height,

✗ KE of the object = 0

✓ KE of the object is the minimum

✗ PE of the object = $\frac{1}{2}mu^2$

✓ PE of the object < $\frac{1}{2}mu^2$

Learn by practice 3

Solve the following problems by considering energy conversion. Neglect air resistance.

1 A ball is projected at an angle of 35° above the horizontal with a speed of 10 m s^{-1} from a position 3 m above the ground. Find the maximum height above the ground that the ball reaches.

2 A fire breaks out in a remote area. A helicopter carrying a bucket of water flies horizontally at a constant velocity of 40 m s^{-1}. The bucket is 30 m above the ground. When the bucket reaches a certain position, water is released and then falls onto the fire. Find the speed of the water when it reaches the fire.

3 Peter throws a ball of mass 2 kg from the top of a cliff at a height of 50 m above the surface of the water. The ball is thrown with an initial speed of 5 m s^{-1} and at an angle of 15° below the horizontal.

(a) Find the speed of the ball when it strikes the surface of the water.

(b) How will the answer in (a) change if the ball is thrown

(i) at an angle of 15° above the horizontal,

(ii) at an angle of 30° below the horizontal?

Ext

☆ HKDSE 2017 Paper 1B Q4

→ This video demonstrates how this question is solved.

Video

DSE practice 1 | Change of speed of a projectile

(a) A stationary object is released to fall freely from position X. Find the distance travelled by the object in the first 5 s. (2 marks)

(b) Then, the object is projected horizontally with speed u at X. Its speed is 56 m s^{-1} after 5 s. Find u. (2 marks)

(c) Now, the object is projected from X with speed u at angle θ above the horizontal. What is its vertical displacement as measured from X when its speed is 56 m s^{-1}? (1 mark)

Solution

(a) Take the downward direction as positive.

Consider the motion in the first 5 s.

$$s = ut + \frac{1}{2}at^2 = 0 + \frac{1}{2} \times 9.81 \times 5^2 \quad \text{1M}$$

$$= 122.6 \text{ m} \approx 123 \text{ m} \quad \text{1A}$$

The distance travelled is 123 m.

(b) Take the gravitational PE at the position of the object after 5 s as zero.

Consider the motion in the first 5 s.

$$KE_1 + PE_1 = KE_2 + PE_2$$

$$\frac{1}{2}mv_1^2 + mgh_1 = \frac{1}{2}mv_2^2 + 0$$

$$\frac{1}{2} \times u^2 + 9.81 \times 122.6 = \frac{1}{2} \times 56^2 \quad \text{1M}$$

$$u = 27.0 \text{ m s}^{-1} \quad \text{1A}$$

(c) 123 m (downwards) 1A

Common mistake
Students may not use energy approach and they may wrongly substitute $v = 56$ m s^{-1} in the equation of motion to find u.

The object has the same mechanical energy at X in parts (b) and (c). Therefore, It always has the same KE at the same height in both cases.

Revision 8 Q27 (p.364)

Exercise 8.2

Neglect air resistance.

1 In a squash game, a ball moves at an angle of 20° above the horizontal just after it is hit. It reaches a wall 2 m away after 0.2 s. Find the horizontal speed when the ball reaches the wall.

A 1.96 m s^{-1} B 6.94 m s^{-1}

C 10 m s^{-1} D 15 m s^{-1}

2 A penguin jumps off an iceberg horizontally into the water (Fig a). Which of the following quantities of the penguin determines the time that it takes to reach the water?

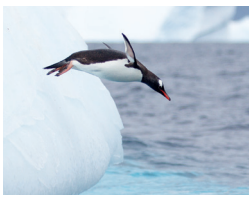


Fig a

A Mass

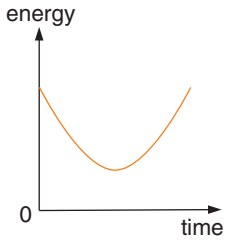
B Initial horizontal speed

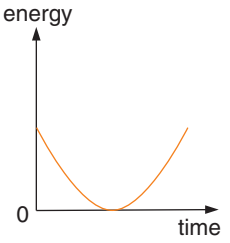
C Initial height above the surface of the water

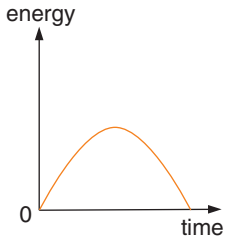
D All of the above

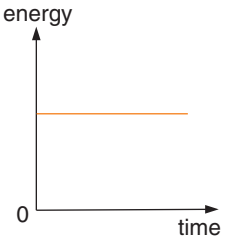
Ext

*3 A ball is projected at a certain angle above the horizontal. Which of the following graphs best shows the variation of the KE of the ball with time?

A


B


C


D


*4 An object is projected horizontally at a speed u from a height h (Fig b). It travels a horizontal distance d and lands after time t.

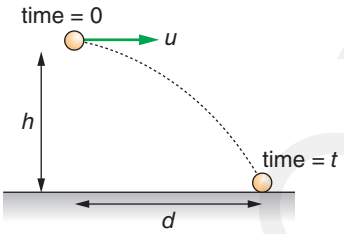


Fig b

Which of the following statements is/are correct?

(1) d increases if u increases.

(2) t increases if u increases.

(3) t increases if h increases.

A (1) only B (2) only

C (1) and (3) only D (1), (2) and (3)

*5 An object is projected at an angle to the horizontal at time $t = 0$. Its velocity at $t = 2$ s is as shown (Fig c).

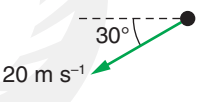


Fig c

What is the initial vertical speed of the object?

A 9.62 m s^{-1}

B 10 m s^{-1}

C 17.3 m s^{-1}

D 19.8 m s^{-1}

*6 Objects X and Y are projected with the same initial speed from level ground. The angle of projection of X is larger than that of Y. Which of the following statements must be correct?

(1) X reaches a higher maximum height than Y.

(2) X has a longer time of flight than Y.

(3) X has a greater range than Y.

A (1) only B (1) and (2) only

C (2) and (3) only D (1), (2) and (3)

7 A toy gun shoots a bullet horizontally at 1.6 m s^{-1}. After travelling a horizontal distance of 0.8 m, the bullet reaches the ground (Fig d).

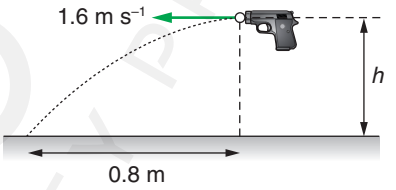


Fig d

(a) How long does the bullet take to land?

(b) Find h.

8 Water is ejected horizontally at speed u (Fig e).

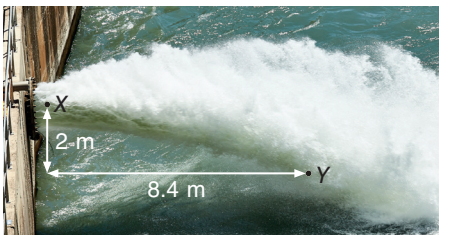


Fig e

(a) How long does the water take to travel from X to Y?

(b) Find u.

9 Sam takes off for a long jump with a velocity of 8 m s^{-1} at an angle of 40° to the horizontal (Fig f). Neglect Sam's size.



Fig f

(a) Find the maximum height that Sam reaches.

(b) Find Sam's time of flight.

(c) Find the range of the jump.

Ext

10 The dotted line in Figure g is the trajectory of a projectile. If it is projected with a higher speed in the same direction, how will its trajectory look? Sketch the new trajectory in the same figure with a solid line.

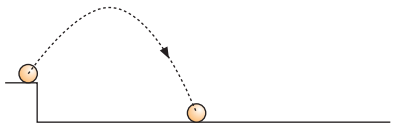


Fig g

***11** Stephen passes a ball horizontally at 10 m s^{-1}. Kevin who is at a horizontal distance of 3 m catches the ball at a position 1.2 m above the ground (Fig h).

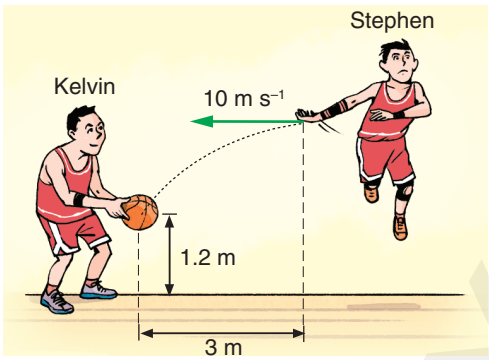


Fig h

(a) Find the height of the ball when Stephen passes it.

(b) Find the speed of the ball when Kelvin catches it.

***12** An athlete throws a javelin at 25 m s^{-1} from a height 1.8 m above the ground (Fig i). Assume that the size of the javelin is negligible. Find the range of the javelin when the angle of projection is 45°.



Fig i

***13** A toy gun aims at a target as shown (Fig j). When the gun fires, the target falls from rest. The target is hit by the bullet after falling 0.5 m. Find the initial speed of the bullet.

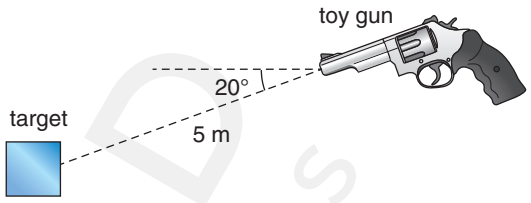


Fig j

***14** A baseball is hit and flies away at 30 m s^{-1} at an angle of 60° above the horizontal (Fig k). It is 0.7 m above the ground when it is hit. Determine whether the ball can fly over the wall to reach the spectator seats. Show your calculation.

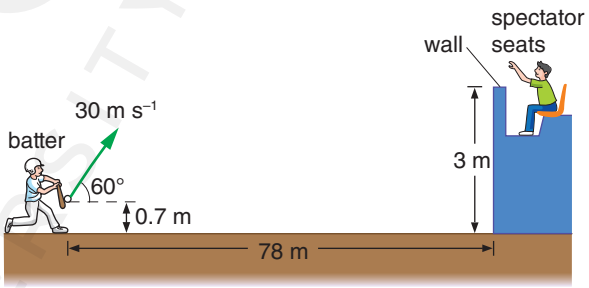


Fig k

***15** A cargo plane moves horizontally at 200 m s^{-1} (Fig l). It releases a package at regular intervals of 1 s. It takes 15 s for a package to fall from the plane to the ground.

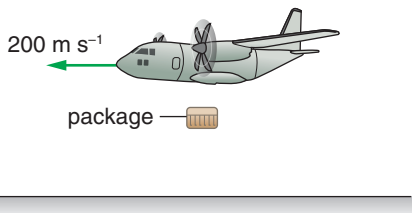


Fig l

(a) When the first package hits the ground, where is the plane relative to the impact point?

(b) How far apart are the successive impact points of the packages on the ground?

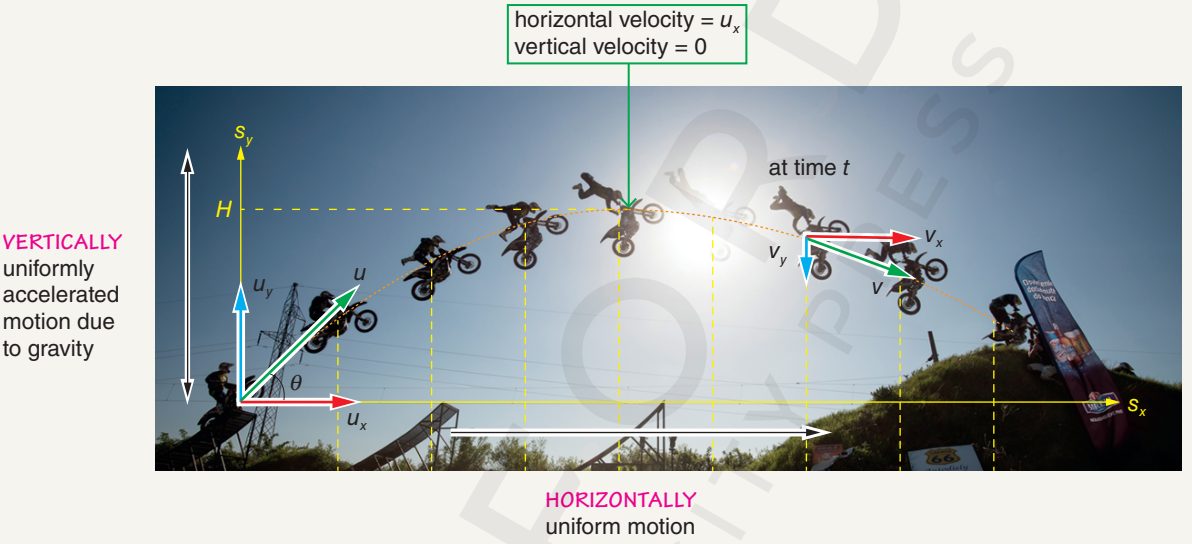
(c) In Figure l, sketch the trajectory of the package.

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Ext

Key concepts 8

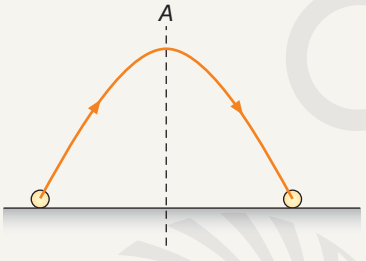
1 Projectile motion



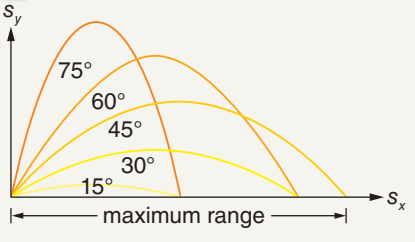
VERTICALLY
uniformly accelerated motion due to gravity

HORIZONTALLY
uniform motion

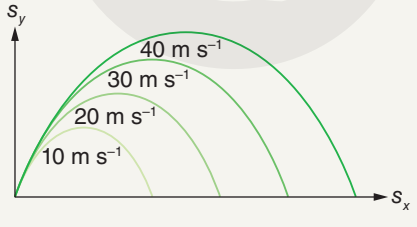
2 A special case: landing level same as launching level



- symmetric trajectory
- time of upward flight = time of downward flight



- $0° < \theta < 45°$: $\theta \uparrow \Rightarrow$ range $\uparrow$
- $45° < \theta < 90°$: $\theta \uparrow \Rightarrow$ range $\downarrow$
- $\theta = 45°$: max. range



- When θ is fixed: $u \uparrow \Rightarrow$ range $\uparrow$

Ext

	Initial value
Horizontal	$u_x = u\cos\theta$ $a_x = 0$
Vertical	$u_y = u\sin\theta$ $a_y = -g$

+ ↑

Apply equations of motion

Velocity	Displacement
$v_x = u\cos\theta$	$s_x = (u\cos\theta)\,t$
$v_y = u\sin\theta - gt$ $v_y^2 = (u\sin\theta)^2 - 2gs_y$	$s_y = (u\sin\theta)t - \frac{1}{2}gt^2$

Maximum height: the vertical distance between the highest point and a reference level

Time of flight: the time interval from projection to landing

Range: the horizontal distance travelled from projection to landing, equal to $u_x \times$ time of flight

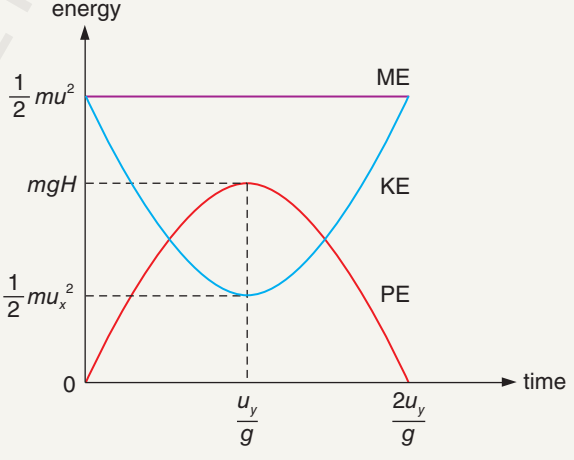
3 Energy change

Assume that there is no energy loss during the conversion between KE and PE.

- mechanical energy (ME)
  = KE + PE = constant
- loss (gain) in KE = gain (loss) in PE
- at the highest point with height H

  min. KE $\left(= \frac{1}{2}mu_x^2\right)$

  max. PE $(= mgH)$



Key terms

1	angle of projection 投射角	p.343
2	parabola 拋物線	p.330
3	projectile 拋體	p.330
4	projectile motion 拋體運動	p.330
5	range 射程	p.344
6	time of flight 飛行時間	p.344
7	trajectory 軌跡	p.330

Ext

Revision 8

Neglect air resistance.

Multiple-choice questions

★1 8.2 A man throws a hammer at a sports event (Fig a). When the hammer leaves his hands at point X, it moves at an angle of 40° above the horizontal. It reaches the maximum height after 1.5 s. Find its speed at X.



Fig a

A $14.4\ \text{m s}^{-1}$ B $17.2\ \text{m s}^{-1}$

C $19.2\ \text{m s}^{-1}$ D $22.9\ \text{m s}^{-1}$

★2 8.2 An object is projected upwards at an angle on level ground. If its initial speed is v, its range is R. Find the range of the object if its initial speed is $2v$.

A $1.4R$ B $2R$

C $2.8R$ D $4R$

★3 8.2 A cat jumps horizontally with a speed u from point X to point Y (Fig b).

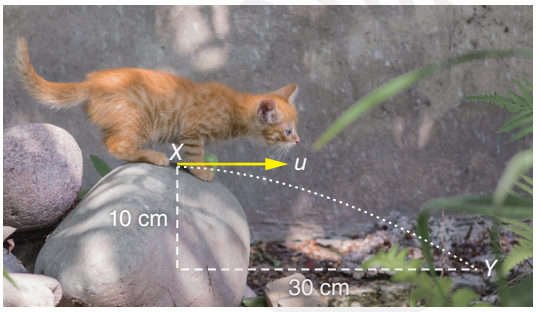


Fig b

Assume that the size of the cat is negligible. Find u.

A $0.48\ \text{m s}^{-1}$ B $2.10\ \text{m s}^{-1}$

C $14.7\ \text{m s}^{-1}$ D $21.0\ \text{m s}^{-1}$

★4 8.2 'The tarik' (Fig c), or pulling tea, is popular in countries like India, Malaysia and Singapore. Suppose the tea leaves the upper jar horizontally with a speed of $0.5\ \text{m s}^{-1}$. Which of the following are the possible horizontal distance s_x and vertical distance s_y between the two jars so that the lower jar can catch the tea? Neglect the sizes of the jars.



Fig c

	s_x / m	s_y / m
A	0.10	0.39
B	0.20	0.39
C	0.10	0.78
D	0.20	0.78

★5 Misc Two objects, P and Q, are projected from the same level at the same time. P is projected vertically upwards at a speed of $30\ \text{m s}^{-1}$ while Q is projected at an angle of 60° to the vertical at a speed of $60\ \text{m s}^{-1}$ (Fig d). The objects land on positions at the same level.

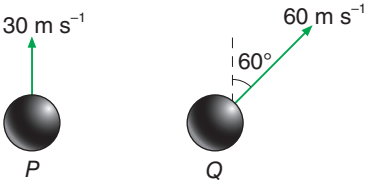


Fig d

Which of the following statements are correct?

(1) P and Q land at the same time.

(2) P and Q have the same acceleration throughout their flights.

(3) P and Q are momentarily at rest when they reach their maximum height.

A (1) and (2) only B (1) and (3) only

C (2) and (3) only D (1), (2) and (3)

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Ext

★6 Misc Objects X and Y are projected from the same position P and land at the same position Q. X is projected at an angle below the horizontal while Y is projected horizontally. Which of the following statements about the objects is/are correct?

(1) They have the same time of flight.

(2) They have the same landing speed.

(3) They have different initial horizontal speeds.

A (3) only

B (1) and (2) only

C (2) and (3) only

D (1), (2) and (3)

★7 8.2 In which direction should a ball be thrown so that its speed is the highest when it reaches the wall (Fig e)? The ball is thrown at a fixed speed and reaches the wall before hitting the ground.

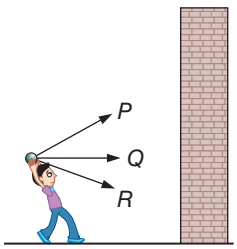


Fig e

A P

B Q

C R

D Same speed in all the above cases

★8 8.2 Streams of water from two hoses meet at point X (Fig f). Suppose each stream of water is ejected at 5.8 m s^{-1} at an angle of 70°. The hoses are 3 m apart. Find the height of X above the ground.

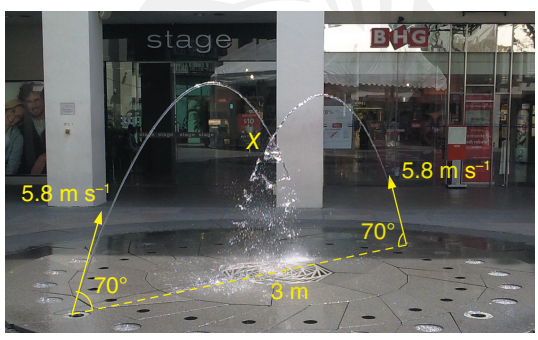


Fig f

A 0.49 m B 1.32 m

C 1.58 m D 1.69 m

★9 8.2 A motorcyclist attempts to jump over a number of cars as shown (Fig g). Each car is 2 m wide. If he leaves the ramp at 30 m s^{-1} and the ramp is inclined at 40° to the horizontal, how many cars can he jump over? Assume that he takes off and lands at the same level, and the size of the motorcycle is negligible.

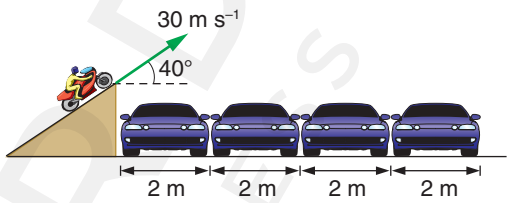


Fig g

A 22

B 45

C 67

D 90

★10 Misc Zoe holds a ball and moves with a constant velocity of 2 m s^{-1} towards the right on a skateboard along a horizontal road (Fig h). The ball is 1.2 m above the ground. Zoe releases the ball when she reaches point P.

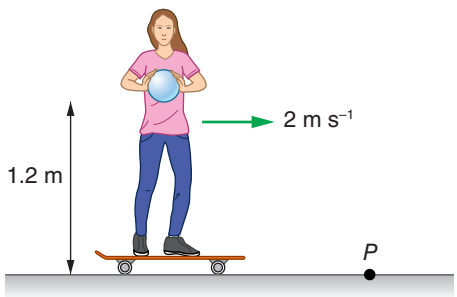


Fig h

Which of the following statements is/are correct?

(1) The ball hits the ground at P.

(2) The ball takes 0.6 s to reach the ground.

(3) The speed of the ball is 5.25 m s^{-1} when it hits the ground.

A (3) only

B (1) and (2) only

C (2) and (3) only

D (1), (2) and (3)

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Ext

**11 8.2 A ball is released from rest at point X on a smooth track (Fig i). It undergoes projectile motion after leaving the track at Y, and eventually falls onto the horizontal ground.

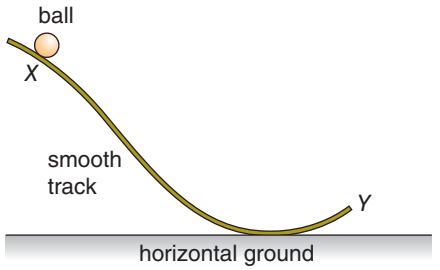


Fig i

Which of the following statements about the projectile motion of the ball must be correct?

(1) The maximum height reached by the ball is lower than X.

(2) The ball's range is related to its angle of projection at Y.

(3) The ball's speed when it hits the ground is related to its angle of projection at Y.

A (2) only B (1) and (2) only

C (1) and (3) only D (1), (2) and (3)

Refer p.352

12 HKDSE 2013 Paper 1A Q13 8.2

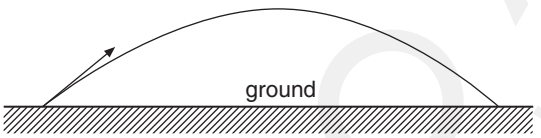
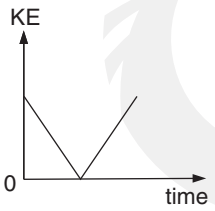


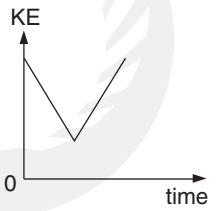
Fig j

A particle is projected into the air at time $t = 0$ and it performs a parabolic motion before landing on the ground as shown. Which graph represents the variation of the kinetic energy (KE) of the particle with the time before landing? Neglect air resistance.

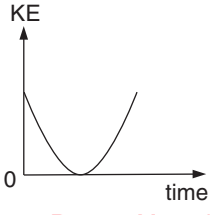
A (14%)



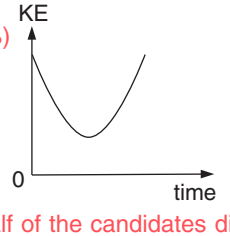
B (8%)



C (43%)



D (35%)



Q12 Exam Report: More than half of the candidates did not realize that the particle still has kinetic energy when it flies horizontally at the highest point.

13 HKDSE 2016 Paper 1A Q10 8.2

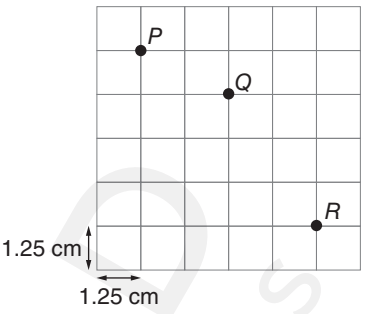


Fig k

The above stroboscopic picture shows a particle projected horizontally at position P into the air in a vertical plane. Subsequently the particle reaches positions Q and R such that the time interval between P and Q is equal to that between Q and R. Each square of the grid measures 1.25 cm × 1.25 cm. Find the particle's speed of projection at P. Neglect air resistance. (g = 9.81 m s^{-2})

A 0.3 m s^{-1}

B 0.4 m s^{-1}

C 0.5 m s^{-1} (51%)

D 0.6 m s^{-1}

14 HKDSE 2017 Paper 1A Q9 8.2

Marbles P and Q of the same mass are shot horizontally. They hit the horizontal ground at points R and S respectively as shown. Neglect air resistance.

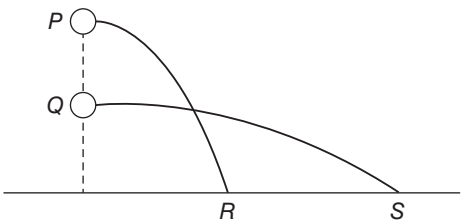


Fig l

Which of the following statements is **incorrect**?

A The initial speed of marble P is smaller than that of marble Q.

B The time of flight of marble P is shorter than that of marble Q. (68%)

C The potential energy loss of marble P is greater than that of marble Q.

D The acceleration of marbles P and Q is the same during the flight.

Ext

15 HKDSE 2019 Paper 1A Q6
8.2

A small ball after projection moves under the effect of gravity only. Its velocity at a certain instant is shown below. What is the speed of the ball **1 s before**? Neglect air resistance. (g = 9.81 m s^{-2})

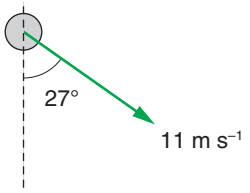


Fig m

A 19.1 m s^{-1} B 9.8 m s^{-1}
C 5.0 m s^{-1} (60%) D 0.2 m s^{-1}

Conventional questions

★**16** A ball is projected at an angle above the horizontal at time t = 0. Its positions at t = 0, T and 2T are as shown (Fig n).
8.2

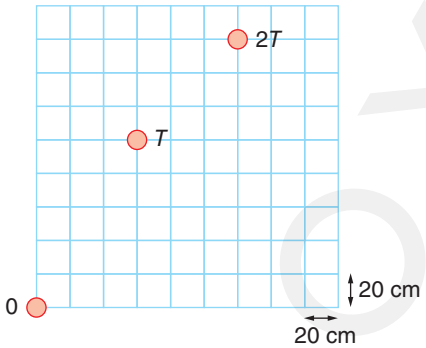


Fig n

(a) Find T. (3 marks)

(b) Find the ball's speed at t = 0. (3 marks)

★**17** A transport helicopter flies horizontally at a constant velocity of 20 m s^{-1} at a height of 125 m (Fig o). It releases a package to a target site on the ground.
Misc





Fig o

(a) How long does it take for the package to fall from the helicopter to the ground? (2 marks)

(b) Find the horizontal distance of the helicopter from the target site when the package is released. (2 marks)

(c) (i) The package is released at time t = 0. Express the vertical distance d from the ground in terms of t.

(ii) Sketch a graph of d against t during the flight of the package. (3 marks)

★**18** A bomber flies horizontally at a constant speed and drops bombs as shown (Fig p).
Misc



Fig p

(a) Suppose the bomber drops 4 bombs consecutively at one-second intervals. The first bomb is dropped at time t = 0. Figure q shows the positions of the bomber using green circles (○) and the bombs using red crosses (×) at t = 0 and 3 s. Mark the positions of the bomber and the bombs at t = 1 s and 2 s. Hence, sketch the trajectory of each bomb. (3 marks)

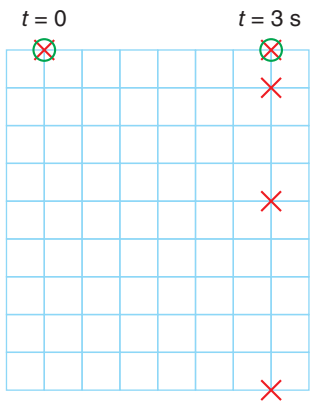


Fig q

(b) If the speed of the bomber is 100 m s^{-1}, find the speed of the first bomb at t = 3 s. (3 marks)

Ext

★19 8.2 A ball is released from rest at point A and moves down along a smooth track ABC as shown (Fig r). The ball then leaves the track at C at 40° above the horizontal.

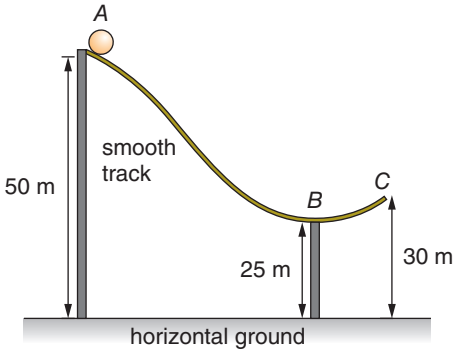


Fig r

Assume there is no energy loss during the whole process.

(a) Find the time of flight of the ball after leaving the track. (4 marks)

(b) Find the range of the ball after leaving the track. (2 marks)

(c) Find the speed of the ball when it reaches the ground. (2 marks)

★20 8.2 In shot-put, an athlete throws a heavy ball from a height of 1.8 m with an angle of projection of 40° (Fig s). His result is 12.0 m. The initial speed of the ball is u and its time of flight is T.



Fig s

(a) Express u in terms of T by considering the horizontal motion. (2 marks)

(b) Find u and T. (3 marks)

(c) Sketch a graph showing how the PE of the ball changes over time during its flight. Take the PE at the ground as zero. (2 marks)

★21 8.2 A skier of mass 70 kg participates in ski jumping. He slides down a ramp with an initial speed of 1.5 m s^{-1} at X and takes off at Y with a velocity of 38 m s^{-1} at 8° below the horizontal (Fig t). He lands on the slope at a horizontal distance 200 m from Y. The work done against friction when he slides from X to Y is 4000 J.

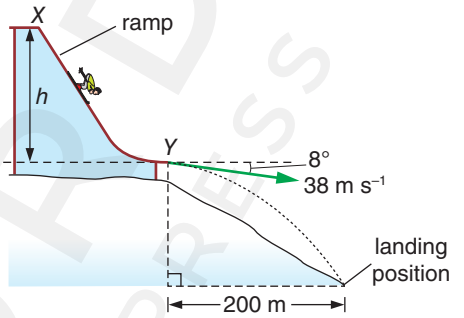


Fig t

(a) Find the vertical distance between X and Y. (2 marks)

(b) Find the vertical distance between the landing position and Y. (3 marks)

★22 8.2 Kelvin shoots a basketball. The ball leaves his hand from a position 2 m above the ground at time $t = 0$ (Fig u). The initial speed of the ball is 5.5 m s^{-1}. The ball passes A and B then into the basket at C at $t = T$. B is the highest point of the ball's trajectory. A and C are at the same horizontal level. The ball's speed at A and B are 3.11 m s^{-1} and 1.88 m s^{-1} respectively.

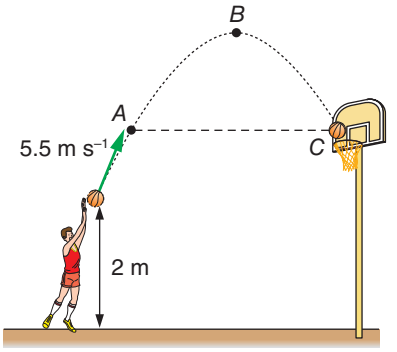


Fig u

(a) What is the speed of the ball at C? (1 mark)

(b) Find the height of B from the ground. (2 marks)

(c) Find the angle of projection of the ball (2 marks)

(d) Find T. (3 marks)

Ext

**23 8.2 In a volleyball game, player A hits the ball at point P and player B hits the ball back at point Q (Fig v). Figure w shows how the speed of the ball varies with time when it travels from P to Q.

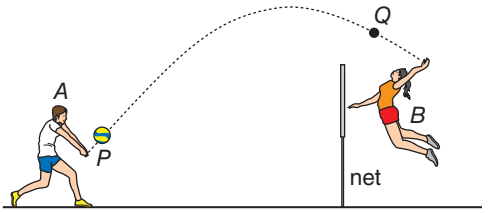


Fig v

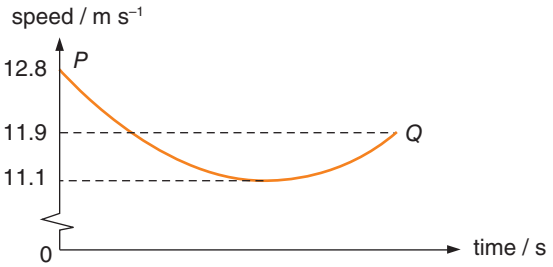


Fig w

(a) Find the angle of projection of the ball at P. (2 marks)

(b) Find the vertical distance between P and Q. (2 marks)

(c) Find the horizontal horizontal between P and Q. (3 marks)

Refer p.350, 351

**24 8.1 You are given an electromagnet, an iron can and a toy gun (Fig x). The electromagnet is fixed at a certain height above the ground. It will be turned off when the toy gun fires. Using the above apparatus, describe an experiment to compare the vertical motion of a projectile with free fall motion. Your description should include the result and conclusion expected. (6 marks)

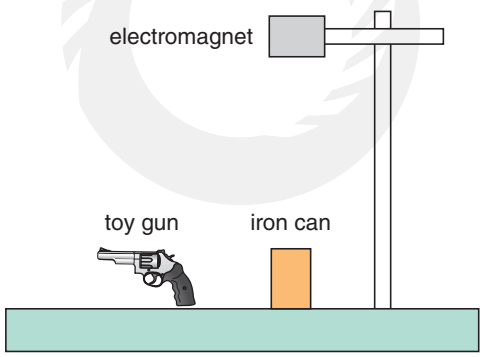


Fig x

Refer p.332, 333

**25 8.2 An astronaut on the Moon throws a stone from the top of a cliff with an initial velocity 10 m s^{-1} at 30° above the horizontal. The initial position P of the stone is 30 m above the ground. After time T, the stone strikes the ground at point Q (Fig y). The acceleration due to gravity on the Moon is 1.62 m s^{-2}.

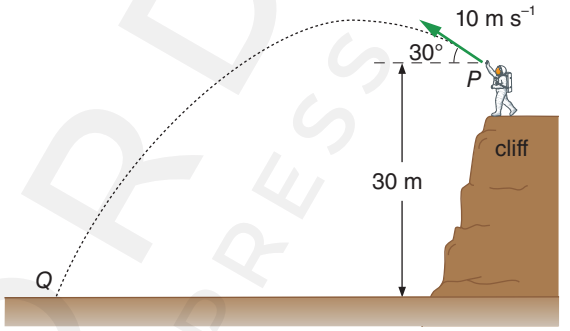


Fig y

(a) Find T. (2 marks)

(b) Find the horizontal distance between P and Q. (2 marks)

(c) Find the speed of the stone when striking the ground. (2 marks)

(d) Sketch the path of the stone in Figure y if the throw is done on the Earth with the same initial velocity. (1 mark)

Refer p.343

**26 Misc A spring gun is installed on a cart which moves with a constant velocity on a smooth horizontal track. The gun points vertically upwards and fires a ball at a certain instant (Fig z).

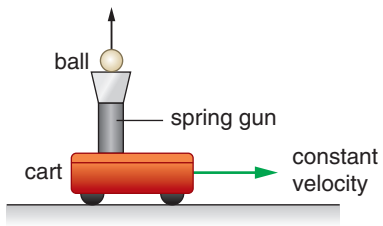


Fig z

(a) Does the ball fall back onto the cart? Explain briefly. (2 marks)

(b) How will the time of flight and range of the ball change if

(i) the cart moves faster;

(ii) the ball is launched at a higher speed? (4 marks)

Ext

(c) The cart is now pulled by a constant force (Fig aa). Can the ball fall back onto the cart after it is fired? Explain briefly. (3 marks)

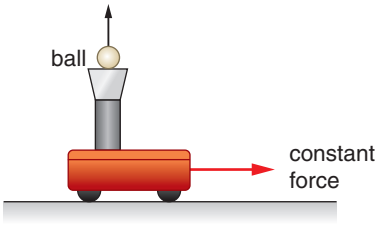


Fig aa

Refer p.345

27 HKDSE 2017 Paper 1B Q4

Misc

(a) A steel ball bearing is released from rest at time $t = 0$. A stroboscopic photo is taken at 0.05 s time intervals. The results are shown in Figure ab. Neglect air resistance.

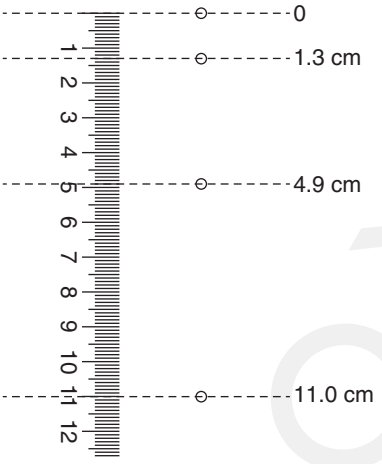


Fig ab

(i) Estimate the acceleration due to gravity using the data in Figure ab. (2 marks)

9.78 m s^{-2}

(ii) The bearing is now projected horizontally instead of released from rest. The bearing is projected at time $t = 0$, and a stroboscopic photo is taken at 0.05 s time intervals. The first and the last image of the stroboscopic photo are shown using circles (○) in Figure ac. For reference, the stroboscopic photo of the bearing released from rest is also shown in the figure using crosses (×).



Fig ac

(1) In Figure ac, mark the positions of the projected bearing in the stroboscopic photo using circles (○). (2 marks)

(2) Given that the bearing is projected horizontally with an initial speed of 1 m s^{-1}, use the results of (a)(i) to calculate the speed of the projected bearing when the last image was taken. 1.78 m s^{-1} (3 marks)

(b) If a small ball is released from rest from the top of a cliff, the speed of the ball becomes constant after a period of time. By considering the forces acting on the ball and using Newton's laws of motion, explain why the speed of the ball becomes constant. (3 marks)

28 HKDSE 2018 Paper 1B Q4

8.2

Figures ad and ae show the cross-section of a fixed spring gun fitted with a small cannon ball.

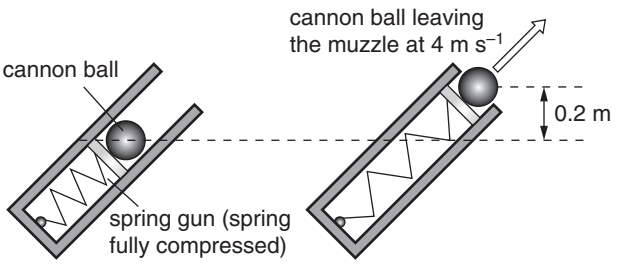


Fig ad

Fig ae

The fully compressed light spring (Fig ad) is released so that the cannon ball of mass 0.3 kg leaves the muzzle of the gun at a speed of 4 m s^{-1} (Fig ae). Neglect air resistance. (g = 9.81 m s^{-2})

employed $v^2 = u^2 + 2as$ to compute the resultant speed of the projected bearing.

Q27 Exam Report: This question tested candidates' knowledge and understanding on projectiles. In general, candidates' performance was fair.

(a) (i) Most answered correctly.

(ii) Some failed to distinguish the horizontal uniform motion and the vertical uniformly accelerated motion of a projectile. Weaker ones even

(b) This part revealed that quite a number of the candidates had misconceptions about Newton's third law of motion — taking weight and air resistance as an action-and-reaction pair in this case.

Ext

(a) The cannon ball is projected at 50° to the horizontal initially and reaches a point horizontally away from the muzzle of the gun at a distance R. Find R and the time of flight t_f of the cannon ball to that point. (4 marks)

(b) If the projection angle is increased to slight greater than 50° while the initial speed remains unchanged at 4 m s^{-1}, explain, **without any calculation**, whether there is any change in t_f. (2 marks)

29 HKDSE 2021 Paper 1B Q3

8.2

A volleyball player serves by hitting the ball from rest at a height of 1.0 m above the end line of the court. The initial speed of the ball is 13.4 m s^{-1} making an angle of 55° with the horizontal. It moves in a vertical plane perpendicular to the end line and finally reaches point C on the opposite end line as shown in Figure af. Neglect the size of the ball and air resistance. (g = 9.81 m s^{-2})

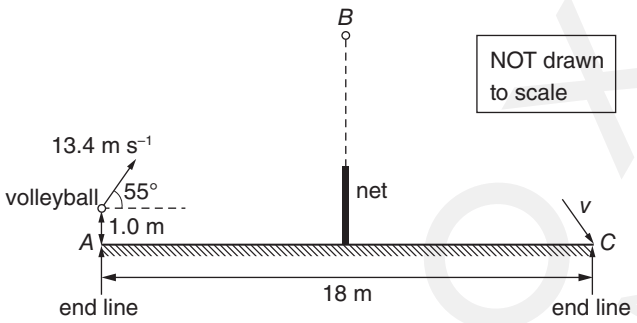


Fig af

Q28 Exam Report:

(a) This part was well answered. However, a few candidates had difficulties in resolving the initial velocity of the ball correctly into its vertical and horizontal components.

(b) Only the more able ones reasoned without any numerical calculation that t_f would increase as the initial vertical velocity v sin θ increased.

(a) (i) The mass of the volleyball is 0.22 kg. Find the work done on the ball by the player. (2 marks)

(ii) Determine the speed v with which the ball hits point C on the ground. (2 marks)

(b) The length of the court AC is 18 m and the net is positioned midway between A and C. It takes time t for the ball to reach point B which is vertically above the net.

(i) State whether the ball is ascending, flying horizontally or descending at B. (1 mark)

(ii) Find t. (2 marks)

(c) Another player suggests that the volleyball can reach point C in a shorter time if it is served with a similar initial speed but at a smaller angle with the horizontal (e.g. 13.2 m s^{-1} at an angle of 35°). Without doing any calculation, explain whether this suggestion is justified. (2 marks)

(d) Volleyball players have to jump and land frequently in a game. Referring to principles of mechanics, explain why volleyball courts with wood rather than concrete flooring may help to protect the players from injuries. (2 marks)

Q29 Exam Report: This question tested candidates' knowledge and understanding of projectile motion in the context of a volleyball game. The overall performance was good.

(a) (i) Most candidates knew that the work done on the ball became its kinetic energy. A few misinterpreted the situation and also considered the ball's gravitational potential energy in the calculation.

(ii) Nearly half of the candidates computed the horizontal and vertical components of the velocity separately rather than simply using the method of conservation of mechanical energy.

(b) (i) Weaker ones held that B was the highest point of the trajectory.

(c) Most knew that the suggested way would result in a shorter time of flight. However, some failed to point out explicitly that this was due to a larger horizontal component of the velocity.

(d) This part was well answered, although some candidates wrongly stated that a concrete floor providing larger friction was the reason.

Ext

Ts may tell Ss that because of the horizontal component of the air resistance, the horizontal velocity of the projectile keeps decreasing.

During the upward flight, the projectile is acted on by a larger downward force ($mg + f \sin \theta$), and so the maximum height reached is lower.

During the downward flight, the projectile is acted on by a smaller downward force ($mg - f \sin \theta$). This makes the speed of the projectile lower than it is at the same level during the upward flight.

Know more

Projectile with air resistance (p.348)

In practice, most projectile motions are affected considerably by air resistance. The air resistance always acts in a direction opposite to the motion of the projectile (Fig a). Moreover, the higher the speed of the projectile, the larger the air resistance.



Fig a *Air resistance always acts in direction opposite to v.*

In the presence of air resistance, a projectile has

- an asymmetric trajectory,
- a reduced maximum height and range (Fig b).

Throwing a shuttlecock and a tennis ball can demonstrate the trajectories when the effect of air resistance is significant and negligible.

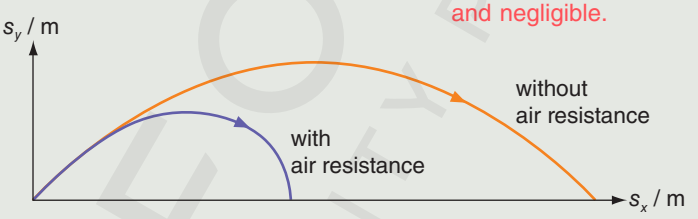


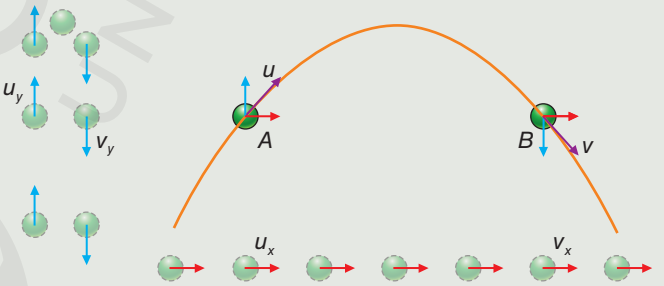
Fig b *Trajectories with and without air resistance.*

Know more

Proofs for some properties in projectile motion (p.348)

Proof: velocity making the same angle with the horizontal at the same height

Consider positions A and B which are at the same height on a projectile's trajectory.



The horizontal component of velocity at A and B is the same ($v_x = u_x$) as the projectile has a uniform horizontal speed.

In the vertical direction, the projectile has a uniform acceleration due to gravity.

Consider the vertical motion from A to B ($s_y = 0$).

$v_y^2 = u_y^2 + 2a_y s_y$

$u_y^2 = u_y^2 + 0$

$v_y = -u_y$ or u_y (rejected)

$\therefore$ The vertical component of velocity at A and B has the same magnitude but is opposite in direction.

Since $\tan \theta = \dfrac{\text{magnitude of vertical component}}{\text{magnitude of horizontal component}}$, the projectile's velocity makes the same angle θ with the horizontal at the same height.

Ext

Proof: parabolic trajectory

For an object projected at a speed u at an angle θ above the horizontal, we can obtain an equation relating s_y and s_x as follows.

Take the upward direction as positive.

Consider the horizontal direction.

$$t = \frac{s_x}{u_x} = \frac{s_x}{u\cos\theta}$$

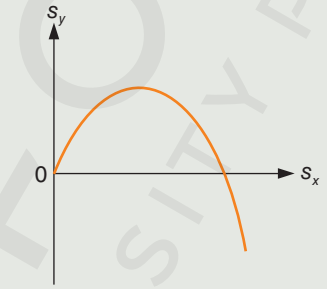
Consider the vertical direction.

$$s_y = u_y t + \frac{1}{2}a_y t^2$$

$$= u\sin\theta \times \frac{s_x}{u\cos\theta} + \frac{1}{2} \times (-g) \times \left(\frac{s_x}{u\cos\theta}\right)^2$$

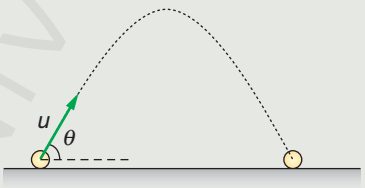
$$= \left(-\frac{g}{2u^2\cos^2\theta}\right)s_x^2 + (\tan\theta)s_x$$

Therefore, the graph of s_y against s_x is a parabola.



Proof: maximum range when $\theta = 45°$

An object is projected with velocity u at an angle θ to the horizontal and lands at the same level.



Take the upward direction as positive.

Consider the vertical motion.

$$s_y = u_y t + \frac{1}{2}a_y t^2$$

$$0 = u\sin\theta \times t + \frac{1}{2} \times (-g) \times t^2$$

$$t = \frac{2u\sin\theta}{g} \quad \text{or} \quad 0 \text{ (rejected)}$$

Consider the horizontal motion.

$$s_x = u_x t = u\cos\theta \times \frac{2u\sin\theta}{g} = \frac{u^2 \times (2\sin\theta\cos\theta)}{g} = \frac{u^2\sin 2\theta}{g}$$

To attain the maximum range,

$\sin 2\theta = 1$

$2\theta = 90°$

$\theta = 45°$

Ext

→ This video demonstrates Expt 8b.

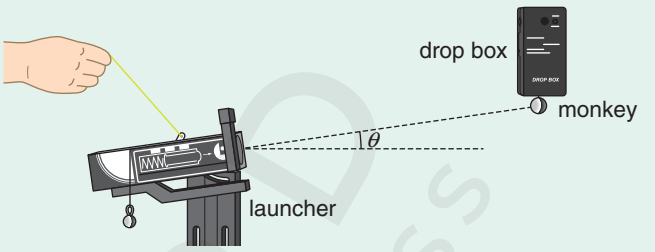
Video

Experiment 8b 'Monkey and hunter experiment (part 2)

1 Set up the 'Monkey and hunter' apparatus as that in Experiment 8a on p.332. Aim the monkey at an angle above the horizontal.

2 Launch a bullet. The monkey starts to fall at the instant the bullet is launched and passes the light-gate. Observe what happens.

3 Repeat step 2 by aiming the monkey at other angles.



Discussion

Does the bullet always hit the monkey?

The same safety precautions as in Experiment 8a on p.332 should be taken.

Know more

Explanation for the result of the 'Monkey and hunter' experiment

In Experiment 8b the bullet always hits the monkey regardless of the tilting angle of the launcher. The result can be explained as follows.

Consider an imagined situation in which gravity disappears. The bullet will move in a straight line and the monkey will stay in the same position. The bullet will hit the monkey and its vertical displacement at different time t is shown below (Fig a).

Now consider the real situation in which gravity exists. Both the bullet and the monkey will move freely under gravity. Gravity causes the bullet to be displaced downwards from the straight-line path. The vertical distances dropped by the bullet and the monkey (highlighted in red in Figure b) due to gravity are the same. Therefore, the bullet will still hit the monkey at the same time (the time is determined by the horizontal motion and is not affected by gravity).

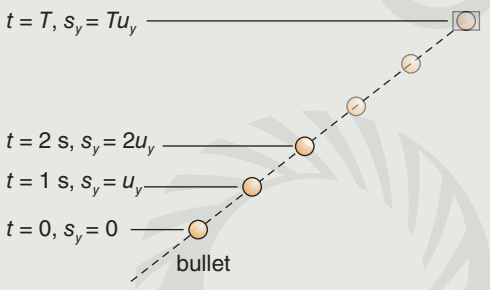


Fig a *Imagined situation (gravity disappears).*

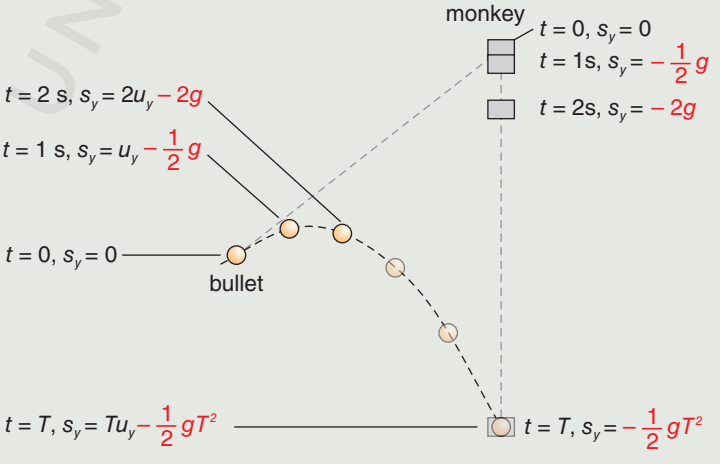


Fig b *Real situation (gravity exists).*